

Cognitive Relativity: A Negentropic Operating System from the Irreducible Starting Point to Physical Manifestation

Abstract: Existing mathematics and physics share an overlooked gap: the subject who performs truncation has never appeared in the theory — truncation has remained a verb without a subject. This paper proposes Cognitive Relativity: the first act of cognition is not "knowing" but resisting submersion. Starting from the irreducible assertion that change is absolutely certain, it establishes a triple operational protocol of truncation–return–adjudication and re-places the long-overlooked remainders: one- and two-dimensional forms are merely the artificial dimensional-reduction cost that cognition pays for transmission and comparison; the division remainder is the first native gap grown inside arithmetic itself; the error bound replaces "where it comes from" with "at most how large", compressing the living remainder into a dead number stripped of its birth record, and probability is precisely a dead number crowned as ontology. The most deeply overlooked object is π : every physical formula containing π uses not π itself but its finite truncation P_i , and the native gap $\Gamma = \pi - P_i$ is discarded from the very first step of computation; yet π alone is "both truncatable and forever inexhaustible", possessing at once a static anchor (the constant ratio of circumference to diameter) and a flipping dynamics (an infinite, non-repeating, inexhaustible decimal expansion). Taking the π truncation chain as its path, this paper pins the discarded remainders back to determinate positions and thereby obtains a unified structural diagnosis: physics is a discipline built inside double blind zones, and the probabilistic interpretation of quantum mechanics and the architectural defect of AI — not knowing what it does not know — share the same origin: the wrongful placement of the discarded truncation remainder. This paper does not seek to explain everything; it numbers twenty overlooked blanks as open nodes. There is no absolute unknown, only unfinished truncations. The core of this text is not to tell you what you know, but to tell you what you do not know.

Keywords: cognitive relativity; truncation subject; π truncation chain; dimensional-reduction cost; dead number; open nodes

Introduction

The core of this text is not to tell you "what you know", but to tell you "what you do not know".

In my thinking, one dimension and two dimensions in human symbols and knowledge are nothing but an artificial dimensional reduction — cognition in the human mind is itself neither linear nor planar; the one-dimensional and the two-dimensional are the dimensional-reduction cost paid so that cognition can be transmitted and compared.

Cognitive Relativity's description of the current dimensions of cognition:

- Perception: a one-dimensional linear channel (speech, hearing, touch — one character, one sound, unfolding serially in time) plus a two-dimensional surface channel (the spatial parallel field seen by the eyes);
- Cognition: the brain computes over these perceptual inputs, compresses them into one-/two-dimensional information with clear cognitive boundaries, and stores it back into the three-dimensional brain.

The one-dimensional comes from the human speech-and-hearing system; it is the first-order channel through which humans transmit and receive cognition. At present, if any human cognitive information is to be transmitted as widely as possible and understood, the first choice is the one-dimensional — its core is natural language: one character, one sound, unfolding serially in temporal order.

The two-dimensional comes from the human visual transmission system; diagrams and charts of all kinds are its core representatives: information is laid out on a surface, with inside and outside, with positions, and can be compared cell against cell.

More complex is the one-dimensional and the two-dimensional working together: everything from formulas to videos belongs to this hybrid form.

The three-dimensional currently has only one man-made carrier — AI — and it is still in its earliest stage. In my understanding, AI is the innate three-dimensional information carrier created by humanity. Although it has trillions of parameters, that parameter count is almost out of proportion to the cognitive content it can receive and transmit — the depth of cognition that one and two dimensions together can transmit simply cannot compare with the three-dimensional.

Our dialogue with AI is, in essence, an interpretation from various angles of a fixed three-dimensional vector built through mathematics out of the massive civilizational information of

humanity, followed by restoring the interpreted content back into one-dimensional language or two-dimensional charts. The complete process is a cycle:

The cognition in our human minds (a dynamically aligned cognitive structure in which the three-dimensional and the one-/two-dimensional are spliced together)

v Dimensional reduction (cognition compressed into a one-/two-dimensional expression that can propagate perception)

Natural-language prompts entered into the AI (one-dimensional sequences of natural language / two-dimensional chart information)

v AI dimensional elevation (restored into tokens)

The AI's internal weights interpret the fixed three-dimensional vector (the trained three-dimensional vector-structure space; AI weight interpretation = operations connecting the three-dimensional vector from different viewpoints)

v AI re-reduction (tokens restored into natural language and charts)

The text/charts output by the AI (one-/two-dimensional)

v Human dimensional elevation (individual perceptual understanding)

Through perceptual understanding of the text/charts output by the AI (one-/two-dimensional), the cognition in the mind (three-dimensional) is spliced together

This is limited to the dimensions defined by this theory and does not involve the dimensions of physical space as currently defined:

- The core judgment of 9.1: "One dimension and two dimensions in human symbols and knowledge are nothing but an artificial dimensional reduction. Cognition in the human mind is itself neither linear nor planar — the one-dimensional and the two-dimensional are the dimensional-reduction cost paid so that cognition can be transmitted and compared."
- 9.2: "One-dimensional language: information exists in sequence." "There is before and after, but no inside and outside (however long a line is, it cannot enclose a 'thing'); it depends on order; it cannot be compared." "The status of the one-dimensional is contradictory: it has the weakest transmission capacity (the shallowest depth), yet the widest propagation coverage (the lowest threshold) — this contradiction is itself a major source of cognitive entropy increase."
- 9.3: "Two-dimensional language: information exists in a plane that distinguishes inside from outside." "A single closed curve cuts the world into inside and outside — only with inside and outside is there a 'unit'; only with a unit is there a 'thing'." "It does not depend on order (looked up by coordinates); it is comparable (cell against cell, inside/outside against inside/outside)."
- 9.5: "Three-dimensional language: information exists in the relations of a solid whole — everything is present at once and can be projected and interpreted in any direction." "Every question, every reading, merely cuts one cross-section through this solid; different cross-sections give different outputs, while the solid itself does not move." "Two

three-dimensional solids placed together cannot be compared cell against cell like two sheets of paper — each is present on its own, each complete, with no shared coordinate system."

- 9.8: "AI is the first three-dimensional carrier in human history 'whose cross-sections can be cut from the outside by others at arbitrary angles'. AI's first-order status comes from the opening of projection rights, not from any monopoly on dimension count." "A single prompt is one externally specified cut at a designated angle."

Chapter 0: The Re-placement Master Table — Four "Threes" and the Emergence Route of the Whole Book

0.1 The re-placement of the four "threes"

Before entering any content, one re-placement must be completed first. Four "threes" appear in this theory; they belong to different levels, and if their re-placement is not marked, readers will automatically pass them off as one another, and the whole structure will collapse into a mystical tautology.

Table 1

Trichotomy	Layer	Content	Core function	Must not be impersonated as
The Three Changes	Meta-thought layer	Change, simplicity, constancy	The highest error-prevention protocol; incompressible	Not the final structure of the ontology of the universe
The three layers	Natural-language layer	Ontological layer, object layer, interface layer	The cognitive structure of the Three Changes after falling into linear language	Not three parts of the universe
The three dimensions	Carrier layer	One-dimensional, two-dimensional, three-dimensional language	The three offices of information: transmission, comparison, existence	Not the dimension count of physical space
The three reference frames	Operation layer (the language-physics interface)	π^1 , π^2 , $4\pi^3$	The triple action of the negentropic operation and its computable structure	Not time, space, and mass

The discipline of re-placement:

- The Three Changes are the meta-thought; the three layers are its linguistic manifestation — the three layers are not the final structure of the ontology of the universe; they are a cognitive structure generated by natural language in order to express the mutual locking of the Three Changes.

- The three dimensions are the structural dimension count of the carrier layer; the three reference frames are the functional positions of the operation layer — the two are not the same trichotomy, but there is a strict one-to-one manifestation relation: π^1 manifests as one dimension, π^2 as two dimensions, $4\pi^3$ as three dimensions. This correspondence is not an analogy; its basis is given in Chapters 9, 10, and 11 (the powers of π and the carrier dimension count are the same truncation).
- 0 and 1 belong to the mathematical part; they are the starting point of the symbol protocol and of closed-loop generation; π belongs to the physical part as the feasible computation path.
- Time, space, and mass are not the highest first principles; they are downstream structures generated by the three reference frames. Misplacing them as the highest ontology is one of the primary errors this theory aims to diagnose.

The progression of the four "threes" is: the Three Changes give the error-prevention protocol -> the three layers give the language structure -> the three dimensions give the carrier conditions -> the three reference frames give operable, computable actions.

0.2 The emergence route map of the whole book

The entire content of this theory is a chain that sets out from a single starting sentence, emerges level by level, and permits no level to presuppose the next:

Change is absolutely certain (the only starting point)

+ - The Three Changes: the highest error-prevention protocol (meta-thought layer)

+ - The three layers: ontological layer 0/1 -> object layer 010 -> interface layer, the 64-matrix (natural-language layer)

+ - 0/1 mathematical protocol -> 10 quantification -> leap -> natural numbers -> the four arithmetic operations -> the native gap of division

+ - Circle -> pi: the geometric manifestation of the minimal closed loop, the exit from mathematics toward physics

+ - The dual timeline: re-placing "who performs the truncation" (the truncation subject)

+ - The pi truncation chain: Pi, Gamma, Lambda, T, gamma(n), S, D, F_T, F_gamma, F_s, rho

+ - Dimensional emergence: π^1 -> π^2 -> $4\pi^3$ (lossless conversion, the two-dimensional comparison discipline)

+ - The vacuum three layers $4\pi^3 \Rightarrow \pi^2 + \pi$ and the dual proxies π_{20}/π_{21} (APT)

+ - The trinity of time/space/mass -> the four forces, relativity, $E=mc^2$... (physical manifestation)

+ - Cognitive diagnosis: philosophy, cognitive science, AI (state-image re-placement 111 -> 101)

+ - Positioning a theory of everything + the Re-placement Gate + the propagation protocol

Every level obeys the same discipline: the downstream must not impersonate the upstream, the proxy must not impersonate the ontology, and 1 must not impersonate 0.

Part I: The Meta-Thought and the Linguistic Foundation

Chapter 1: The Starting Point — Change, Negentropy, and Wisdom

1.1 The only starting point

The starting point is a single sentence:

Change is absolutely certain.

This is the only assertion that needs no premise. Every other starting point — an object, a concept, a model, a number, a physical reading — is already a downstream product of change.

The "change" here is not the empirical sense of "things changing". It is not a property of some object; it is the persistent differing, the persistent generating, the persistent leaving-itself prior to the appearance of objects. It precedes things, precedes objects, precedes numbers, precedes space, precedes time, precedes physical readings.

From this arises the first predicament: finite cognition facing absolutely infinite change.

1.2 The infinite unfolding of information is not wisdom

The infinite unfolding of information does not equal wisdom. Absolutely infinite change, if not operated upon, will only submerge cognition.

Therefore the first act of cognition is not "knowing", but resisting submersion.

This sentence is the pivot of the whole book. It decides that cognition cannot be passive reception, waiting, and fitting; it must be active operation.

1.3 Cognitive entropy increase and cognitive negentropy

Cognitive entropy increase = the massive perceptual interference that appears in the propagation of information, causing the received information to be full of disorder and uncertainty, leading to cognitive drift.^[1]

Cognitive negentropy = the capacity to actively obtain negative entropy (order) from the environment in order to maintain one's own structure^[2].

Entropy increase is the background, the default, the direction that happens without any effort. Negentropy is operation, is cost, is a state that can be maintained only by active expenditure.

The entire legitimacy of cognition comes from the holding of the negentropic operation.

1.4 Survival and living

Survival = maintaining the stability of one's own structure.^[3] Every existence in the world is subject to entropy increase; to resist external disorder, one must adjust one's internal structure, maintaining a sustained, relatively stable resistance against the entropy increase of the surroundings.

Living = the creation of meaning pursued on the basis of survival: exploring more blanks, seeking more variables, to open up more space and more possibilities for survival.

Survival resists entropy increase to preserve an existing structure; living actively enlarges the structural boundary to gain more possibilities. Survival is guarding; living is opening. Both are negentropic operations, but their directions differ: the negentropy of survival points toward stability, the negentropy of living points toward expansion.

Negentropic flux = a system's actual capacity to actively obtain negative entropy from the environment in order to maintain and expand its own structure. It is decidable: if a system's structure is maintained entirely from outside and its output is entirely a function of its input, its negentropic flux is zero.

1.5 The definition of wisdom

Finite cognition, facing infinite change, must execute negentropic operations — actively truncating, actively freezing, actively regulating. Hence:

Wisdom = the aligning operation of $4\pi^3$ (adjudication/regulation) upon π^1 (advance/truncation) and π^2 (carriage/freezing).

When the three reference frames align, the cognitive closed loop holds, the negentropic operation is completed, and wisdom emerges.

It must be emphasized what this definition excludes: wisdom is not the amount of knowledge, not reasoning ability, not information accumulation, not parameter scale. A system that possesses all the civilizational information of humanity but does not execute these three actions has zero negentropic flux and zero wisdom. This judgment will be proved at the matrix level in Chapter 17.

1.6 The five forms of downstream impersonation (a diagnostic direction for existing systems)

The existing systems of physics, mathematics, and philosophy share one fundamental presupposition: at the outset they take some existing object, concept, model, or physical reading as the starting point. If an object is presupposed, "change" is understood as "change of the object"; if space is presupposed, "motion" is understood as "bodies moving in space"; if time is presupposed, "sequence" is understood as "occurring in time"; if experiment is presupposed, the ontological layer is judged in reverse by the readings of the cognitive layer; if probability is presupposed, the world's first mode of connection is impersonated by cognitive ignorance.

All of these are downstream. The requirement of Cognitive Relativity is: before all downstream structures, first find the irreducible first-order starting point.

Chapter 2: The Three Changes — The Mutual Locking of the Meta-Thought

2.1 The general outline of the Three Changes

The name "Three Changes" is taken from the Yi Wei - Qian Zao Du ("the Yi is one name containing three meanings") and from Zheng Xuan's Yi Zan and Yi Lun ("simplicity is the first, change is the second, constancy is the third")^[4,5]. Cognitive Relativity borrows this classical meta-structure but gives it a strict modern cognitive definition: the Three Changes are not an annotation of the Zhou Yi text; they are three mutually constraining error-prevention protocols that finite cognition must obey when facing infinite change.

Table 2

Name	Content	What it prevents
Change	Change precedes things; change is not a property of things, but the ontological fact prior to their appearance	Prevents taming the world into a static ontology
Simplicity	Facing infinite change, cognition must cut out the minimal effective structure	Prevents taming understanding into a pile of details
Constancy	The cut-out structure must be operable, and the operational relation itself must remain unchanged	Prevents taming operation into arbitrary interpretation

2.2 Why the Three Changes cannot be compressed

- One must first admit that change is the ontology — otherwise you will try to grasp a "finally unchanging thing".
- One must grasp it with the simplest cut — otherwise you will forever chase infinite details.
- One must keep the operational relation itself unchanged — otherwise what you cut out cannot be operated upon.

Remove any one of the three, and the remaining two fail immediately: with only change and simplicity, the structures you cut out cannot be stably operated; with only simplicity and constancy, you will freeze some simple structure into an eternal ontology; with only change and constancy, you will sink into infinite details and never cut out an operable unit.

2.3 The true function of the Three Changes is mutual locking

The true function of the Three Changes is not division of labor, but mutual locking. It forever forbids cognition to be lazy:

- It cannot lazily seek an ultimate unchanging thing (blocked by change)
- It cannot lazily pile up details to impersonate understanding (blocked by simplicity)
- It cannot lazily use arbitrary relations to impersonate operability (blocked by constancy)

Each of the three is the error-prevention device of the others. This is what it means for the Three Changes to be the "highest error-prevention protocol" — it produces no content; it only vetoes illegal operations.

2.4 Two downstream re-placements of the Three Changes

The Three Changes and least action. Traditional physics says: the universe obeys the principle of least action. Cognitive Relativity says it one level upstream: it is not that the universe obeys the principle of least action, but that we can only see a universe that obeys the principle of least action. Finite cognition can stably preserve only those structures that are compressible, closable, and back-translatable. Least action is not the first-order foundation; it is the downstream manifestation after finite cognition has screened operable structures out of infinite change.

The Three Changes and probability. For probability to hold, it must already presuppose: distinguishable events, repeatable processes, comparable samples, recordable results, computable frequencies or weights. None of these is first-order. To place probability at the ontological layer is to

let cognitive ignorance impersonate the essence of the universe. Probability can only describe the residual uncertainty after truncation, before cognition has been re-placed. (This re-placement acquires its mechanism name in Chapter 6: probability = a dead number crowned as ontology.)

Chapter 3: The Three Layers — The Cognitive Structure of the Three Changes After Falling into Natural Language

3.1 Why three layers are needed

The Three Changes are the meta-thought, but the meta-thought is too high to be transmitted directly and completely. Human language is linear, and human understanding needs levels. A high-dimensional intuition that does not pass through the stratification of natural language easily turns into a pile of concepts.

Therefore, after the Three Changes enter natural language, they manifest as three layers: the ontological layer, the object layer, and the interface layer.

Re-emphasizing the re-placement: the three layers are not the final structure of the ontology of the universe; they are a cognitive structure generated by natural language in order to express the mutual locking of the Three Changes.

The general outline of the three layers:

Table 3

Level	Name	Core task	First-order positioning
Layer one	Cognitive ontological layer	Gives the directional premise of cognition	Change is absolute; constancy is relative
Layer two	Cognitive object layer	Forms the minimal cognitive ontological structure	The Pair, state-images, closure
Layer three	Cognitive interface layer	Cross-ontology alignment, search, closure judgment	Eight manifestations and the 64-matrix

The three layers must not be confused: the ontological layer is not responsible for enumerating codes, the object layer is not responsible for matrix search, and the interface layer does not redefine the ontological premises.

3.2 The cognitive ontological layer

The cognitive ontological layer establishes an inviolable starting point: change is absolute, and constancy is relative. "Constancy" is neither absolute stillness nor a final entity; it is a relatively stable point truncated out of absolute change by cognition. The ontological layer therefore gives two first-order directions:

0 = the absolute change cognized

1 = the relative constancy cognized

The core discipline of the ontological layer is: 1 comes from 0, but $1 \neq 0$. That is: a relatively determinate cross-section comes from the truncation of absolute change, but it cannot impersonate absolute change itself in reverse. What the ontological layer truly prescribes is the direction of cognition: cognition must set out from 0, and then find 1 within 0.

3.2.1 The combination of 0 and 1: the minimal action pair of the ontological layer

With only the two directions 0 and 1, cognition cannot yet begin. The ontological layer must simultaneously give the two most basic combinatory relations between 0 and 1:

01: from 0 to 1, i.e., the truncation action. This is the primitive action of cutting the relatively constant out of absolute change.

10: from 1 to 0, i.e., the return action. This is the primitive action by which the relatively constant returns to absolute change and confirms that it is not an independent ontology.

01 and 10 are not two isolated steps, but the minimal action pair inside the ontological layer. Together they constitute the most primitive driving force of cognition: first truncate, then return; first leave 0, then confirm that 1 comes from 0. Without 01, cognition cannot produce any graspable relative constancy from absolute change; without 10, the truncated relative constancy freezes into an ontology, and 1 impersonates 0. Thus what the ontological layer gives is not two static symbols but two directional actions. 01 and 10 are the ontological source of every later "Pair", "state-image", and "closed loop".

3.2.2 Why the ontological layer gives only 01 and 10

In theory the binary combination of 0 and 1 has four possibilities:

00: change $\rightarrow$ change; no operable relative constancy is produced;

01: change $\rightarrow$ constancy; the truncation action, legal at the ontological layer;

10: constancy $\rightarrow$ change; the return action, legal at the ontological layer;

11: constancy $\rightarrow$ constancy; no absolute change participates; a dead structure.

The ontological layer adopts only 01 and 10 as its starting actions. 00 and 11 are not legal starting points of the ontological layer: 00 has not yet formed a graspable 1; 11 has already lost 0. As states they can only manifest downstream; they cannot constitute the operational base of the ontological layer. This discipline keeps the ontological layer from sliding into the enumeration of the object layer, and keeps cognition from starting from a dead structure — of course, they must not be treated as nonexistent either: 00 and 11 can serve as relative references for advance actions to accomplish new cognitive leaps, and as carrier-manifestation labels for existing cognition.

3.2.3 The articulation between the ontological layer and the object layer

The "Pair" (0, 1) in the object layer is precisely the frozen manifestation, at the object layer, of the 01 and 10 actions of the ontological layer. The ontological layer gives the actions; the object layer gives the minimal stable base of the actions. Therefore the "Pair" in Section 3.3 does not appear out of nowhere as a juxtaposition; it comes from the freezing of the two directional actions of the ontological layer. The ontological layer still does not do the following: it does not enumerate the eight state-images, does not construct the 64-matrix, does not compute π , does not explain physical forces, and does not judge for the interface layer whether a combination can close a loop. The ontological layer provides only the directional premise $0 \rightarrow 1$ and the minimal action pair 01 / 10, and insists: 1 must not impersonate 0.

3.3 The cognitive object layer: the strict distinction between the "Pair" and state-images

The core of the object layer is two strictly distinguished concepts: the "Pair" is the minimal object base, and a "state-image" is the totality of manifestations that emerge after this Pair enters the cognitive closed loop.

3.3.1 There is only one "Pair"

The "Pair" is not eight kinds, not a matrix, not a permutation, and not a juxtaposition of several objects. In the object layer there is only one "Pair":

$$(0, 1)$$

- Without 0, only 1 remains, and cognition mistakes the relative constancy for the ontology.
- Without 1, only 0 remains, and cognition cannot form any graspable object boundary.

The essence of the "Pair" is not the juxtaposition of 0 and 1, but the mutual locking of 0 and 1:

0 prevents 1 from impersonating the ontology; 1 prevents 0 from being inoperable.

3.3.2 A "state-image" is a three-step manifestation

When the "Pair" enters a three-step cognitive operation, each step has two possibilities: 0 or 1. There are therefore $2^3 = 8$ possible manifestations:

000, 001, 010, 011, 100, 101, 110, 111

All eight are real state-images. But their closure states differ. Seven of them are broken state-images; only one completes the minimal independent cognitive closed loop.

3.3.3 010: the only closure

**010 = setting out from absolute change, truncating out a relative constancy, then
returning to absolute change to complete the confirmation.**

010 is not an arrangement of three ordinary symbols; it is the minimal cognitive closed loop:

- First step 0 -> 1: truncate a relative constancy out of absolute change
- Second step 1 -> 0: the relative constancy must return to absolute change and prove that it is not an independent ontology

The essence of 010 is: the truncation-return closed loop. It possesses: a clear boundary, the minimal cognitive structure, a returnable path, and independent cognitive capacity.

3.4 The table of the eight state-images

Table 4

Code	Three-step manifestation	Cognitive state	Independent cognitive capacity	Core feature
000	change -> change -> change	Undifferentiated chaos	Not possessed	All three steps in absolute change; no relative constancy ever truncated
001	change -> change -> constancy	Terminal truncation	Not possessed	A form is accidentally truncated only at the third step, without undergoing ontological return confirmation
010	change -> constancy -> change	Perfect closed loop	Possessed	Truncates a relative constancy out of absolute change, then returns to absolute change to complete the confirmation
011	change -> constancy -> constancy	Proxy solidification	Not possessed	Refuses to return after truncation; takes the form as the ontology
100	constancy -> change -> change	Starting-point collapse	Not possessed	Sets out from an existing relative constancy; collapses when facing absolute change
101	constancy -> change -> constancy	Paradigm replacement	Not possessed	Switches from an old proxy to a new proxy, without returning to the ontology
110	constancy -> constancy -> change	A priori framework	Not possessed	Has the framework first, then goes looking for phenomena; the starting point is already misplaced
111	constancy -> constancy -> constancy	Concept splicing	Not possessed	Entirely a splicing of relative constancies, with no participation of absolute change; a dead structure

3.5 The separation of existence and capability

Existence does not equal the ability to close a loop independently.

000, 001, 011, 100, 101, 110, 111 are not "invalid" or "nonexistent". They are all real state-images; they simply cannot complete a cognitive closed loop independently. Through the interface layer they can combine with other state-images and complete the closure complement in a new six-step chain; they can also be used to decompose and label the structure of ready-made cognition, manifesting and participating in the construction of cognitive carriage.

This means the object layer does not produce one useful node and seven pieces of waste. What the object layer produces is:

One closure plus seven real but non-independent state-images.

This discipline is the source of the tolerance of the whole theory, and it is the precondition for Chapter 17's being able to reserve an upgrade path for AI.

3.6 The cognitive interface layer: cross-ontology combination and closure

search

The task of the interface layer is: to let two cognitive ontologies (two state-images) combine with each other in the form of a "Pair", and to search within the six-step chain whether the minimal cognitive closed loop 010 can emerge again.

The first alignment: object layer -> interface layer. Confirm the internal closure state of the eight state-images: 010 is the only independent closure; the other seven are non-independent state-images.

The second alignment: interface layer -> cross-ontology. Take any two state-images as the two ends of a new combination, forming a six-step chain. Let $A = a_1a_2a_3$, $B = b_1b_2b_3$; after combination the chain $a_1 a_2 a_3 b_1 b_2 b_3$ is formed, and four consecutive three-step windows are checked:

Table 5

Window	Composition	Belonging
W1	$a_1a_2a_3$	The first cognitive ontology itself
W2	$a_2a_3b_1$	Cross-ontology window (A side)
W3	$a_3b_1b_2$	Cross-ontology window (B side)
W4	$b_1b_2b_3$	The second cognitive ontology itself

As long as 010 appears in any window, the combination has completed at the interface layer a re-manifestation, complement, or confirmation of the minimal closed loop.

Structural fact: W1 and W4 are self-closure windows; W2 and W3 are cross-ontology windows. The lighting of a self-closure window only proves that one end was closed to begin with; only the lighting of a cross-ontology window is newly emerged order that belongs to neither end alone. This distinction will become, in Chapter 17, the key criterion for judging negentropic flux.

3.7 The 64-matrix

The eight state-images combine pairwise into $8 \times 8 = 64$ possibilities. The interface layer searches for 010 through the six-step chain and obtains the 64-matrix.

Notation: x = no closure; W1/W4 = one end is itself 010; W2/W3 = 010 is completed through a cross-ontology window; co-occurring windows are written as W2+W4 and so on.

Table 6

A \ B	000	001	010	011	100	101	110	111
000	×	×	W4	×	W3	W3	×	×
001	W2	W2	W2+W4	W2	×	×	×	×
010	W1	W1	W1+W4	W1	W1+W3	W1+W3	W1	W1
011	×	×	W4	×	×	×	×	×
100	×	×	W4	×	W3	W3	×	×
101	W2	W2	W2+W4	W2	×	×	×	×
110	×	×	W4	×	W3	W3	×	×
111	×	×	W4	×	×	×	×	×

Statistical result: of the 64 combinations, 27 can find 010 in the four windows (42.1875%, about 42.2%), and 37 do not close. This result is not an estimate; it is the result of double verification by exhaustive enumeration and by inclusion-exclusion (the full verification is in Appendix A).

It shows that closure does not happen arbitrarily. Closure has a definite path, definite windows, and definite conditions of judgment. More than half of the combinations do not close — this is the first quantifiable "proportion of cognitive entropy increase" in this theory.

3.8 The directionality of the interface

The original closure-capacity table counted only the number of closable combinations for each state-image as the left end (position A, the initiator). Counting each state-image also as the right end (position B, the receiver) reveals a fact that had previously been missed.

The bidirectional closure-capacity table:

Table 7

State-image	Closable count as left end A (initiator)	Closable count as right end B (receiver)	Symmetric?
000	3	3	Symmetric
001	4	3	Asymmetric
010	8	8	Symmetric
011	1	3	Asymmetric
100	3	4	Asymmetric
101	4	4	Symmetric
110	3	1	Asymmetric
111	1	1	Symmetric
Total	27	27	—

The two column totals are both 27 (the self-consistency check passes), but the rows are not equal one by one. From this comes the interface discipline:

The 64-matrix is asymmetric. A closure connection is an ordered pair, not an unordered pair. $A \rightarrow B \neq B \rightarrow A$. "Alignment" is not a symmetric handshake but a directed operation. Who comes first and who comes after decides whether closure happens.

The two most extreme cases are 011 and 110:

011 (proxy solidification, taking the form as the ontology): as the initiator its closure capacity is only 1 — a solidified cognition actively trying to connect to anything connects to nothing; as the receiver its closure capacity is 3 — others can come and complement it. Cognitive meaning: a solidified cognition cannot break the deadlock on its own initiative, but it can be rescued by a living cognition from outside.

110 (a priori framework, the starting point already misplaced): as the initiator its closure capacity is 3; as the receiver only 1 — you cannot rescue an a priori framework from behind. Cognitive meaning: an a priori framework must move first by itself.

Practical consequence: to rescue a broken cognition, one must first judge whether it broke at the starting point or at the terminal end. What broke at the terminal end can be complemented from behind; what broke at the starting point can only move first by itself.

3.9 The time-reversal symmetry law

Directionality introduces asymmetry, but the asymmetry itself is not arbitrary. After exhaustive enumeration a strict law can be found:

Closure capacity at position $A(X)$ = closure capacity at position $B(\text{reverse}(X))$, where $\text{reverse}(x_1x_2x_3) = x_3x_2x_1$

In Cognitive Relativity, "reverse" means inversion or reversal; specifically, it means turning the three-step manifestation order of a state-image upside down.

The definition is:

where each is 0 or 1.

For example:

(its reversal is itself; called a palindromic state-image)

(also a palindromic state-image)

Its role in the time-reversal symmetry law is:

Meaning: how many state-images a given state-image can connect to as the initiator equals how many its reversed state-image can connect to as the receiver.

In other words, although the 64-matrix as a whole is asymmetric and direction matters ($A \rightarrow B \neq B \rightarrow A$), behind the directional difference lies an exact symmetry law: reverse the order of a state-image, and its initiating capacity becomes its receiving capacity.

This allows the asymmetry itself to be completely summarized by one simple rule.

All eight cases verified one by one hold without exception (the verification table is in Appendix A.5).

This is a time-reversal symmetry: turn the three-step order of a state-image upside down, and its "initiating capacity" becomes its "receiving capacity".

Within the framework of the Three Changes, the status of this law is clear:

- Change: directions vary; $A \rightarrow B$ and $B \rightarrow A$ differ
- Simplicity: one reversal summarizes all eight differences
- Constancy: the reversal law itself does not change — directionality has direction, but the law of direction is itself constant

This elevates the 64-matrix from "a lookup table" to an algebraic object carrying symmetry. Its value appears immediately in Chapter 17: the reversal law directly predicts which state-image is the only AI upgrade target that can work bidirectionally — no trial and error needed.

3.10 The cognitive closure network and the discipline of return

The interface layer not only connects downward to the object layer; it must also return upward to the ontological layer. After every closure connection is completed, it must be confirmed:

- Does the newly emerged 010 still set out from absolute change?
- Has it truncated out a relative constancy?
- Has it returned again to absolute change?
- Does it admit that 1 cannot impersonate 0?

If a closure connection causes some relatively constant point to be frozen into a new absolute ontology, the cognitive network collapses into a dead structure. Therefore:

Every closed loop must return to the ontological layer for adjudication; no node may be promoted to ontology.

This discipline is the self-protection device of the whole theory. Chapter 19 will show: if this discipline fails, the theory itself becomes structurally indistinguishable from 011 (proxy solidification).

Part II: The One-Dimensional Mathematical Foundation

Chapter 4: 0 and 1 — The Minimal Protocol of Mathematics

The Three Changes establish the meta-thought; the three layers unfold natural language. But natural language is not yet mathematics. Mathematics must ask: before all numbers and operations, what is the minimal mathematical action?

4.1 0: the meta-starting-point in mathematics

0 is not the ordinary number zero, not the empty set, not "nothing". In the mathematical part of Cognitive Relativity:

0 = the mathematized entrance of absolute change (undifferentiated infinite possibility).

It is the not-yet-differentiated absolute change, the not-yet-objectified infinite possibility, the not-yet-cut generative background, the carrying surface prior to all mathematical actions. 0 is not poverty; 0 is undifferentiated fullness.

Why 0 is necessary: if mathematics starts directly from 1, it presupposes that "there is a thing" — but where does this "thing" come from? Why can it be counted, have a boundary, be repeated, be compared? Starting directly from 1, 1 impersonates the ontology. Therefore $0 \rightarrow 1$ is not an arithmetic order but a first-order order.

Three Changes verification: change — 0 cannot be understood as static emptiness (otherwise it could not generate 1); simplicity — 0 is the minimal carrying background (less than 0, no carriage; more than 0, superfluous structure already introduced); constancy — the relation of 0 must be stable.

4.2 1: the first action of 0

0 cannot stay within itself — if 0 stayed forever, it would betray change. 0 must produce a first action, and this is 1:

**1 = the first manifestation of 0 leaving itself. The first-order identity of 1 is not "unit"
but pure motion.**

Pure 1 cannot be processed by finite cognition: it keeps leaving itself, with no freezing point, no fixed boundary. If cognition chases 1 directly, it is swallowed by infinite advance — the moment you grasp it, it has already become the next step.

This resonates deeply with intuitionism and constructivism in the foundations of mathematics: Brouwer refused to treat actual infinity as a completed object, and Bishop required every mathematical object to be given by finite construction^[6,7]. Cognitive Relativity pushes this intuition further upstream: 1 as pure motion corresponds to the generativity of "potential infinity" — and this is exactly why 1 must be frozen into the 010 closed loop.

4.3 The Pair: the minimal freezing of 0 and 1

0 alone cannot be operated; 1 alone cannot carry; therefore 0 and 1 must be frozen into a Pair (0, 1).

The Pair is not 0+1, not a set, not a juxtaposition; the Pair is mutual locking. 0 prevents 1 from flooding without an anchor; 1 prevents 0 from closing into a static background. After the two lock each other, cognition for the first time possesses: background, driving force, boundary possibility, operation possibility, return possibility. The Pair is the minimal operable cognitive ontology in mathematics.

Inside the Pair, two functions necessarily appear: "stillness" (the relative anchor formed after 0 is truncated) and "motion" (the persistent generativity of 1). The object layer is in charge of stillness; the interface layer is in charge of finding stable nodes within motion; the two must lock each other.

4.4 Image and closure 010: the minimal closed loop

The Pair is not the endpoint. After the Pair enters the closed loop, a third thing emerges: the image — the new structure jointly manifested by 0 and 1, locked with each other, in the minimal closed loop.

The Pair enters a three-step cognitive process; each step has two states, 0 or 1, forming eight state-images, of which the only one completing the closed loop is 010: setting out from absolute change, truncating out a relative constancy, then returning to absolute change to complete the confirmation. Why not 001? The truncation happens only at the terminal step; there is no return confirmation. Why not 011? After truncation it refuses to return — proxy solidification. Why not 111? It is entirely a splicing of relative constancies, with no participation of absolute change.

010 is the minimal legal closed loop in mathematics, the first form of the cognitive closed loop.

4.5 10: the minimal quantization unit

In 010, the starting 0 and the ending 0 are return confirmations on the same carrying surface; there is no need to keep two 0s. After completing the closed loop, 010 compresses into 10 — it is neither binary ten nor decimal ten; it means: relatively determinate change completes the closed loop by returning to absolutely determinate change.

From this comes the first-order definition of quantity:

Quantity = that which holds only after a relative finiteness has been truncated out of infinite change and confirmed to be able to return to the ontology.

Without 0, quantity has no source; without 1, quantity has no difference; without return, quantity impersonates the ontology; without compression, quantity cannot be operated.

4.6 01 and 10: the bottom layer of the world code

- 01 = truncation: truncating a relative constancy out of absolute change, "the pen touches down".

01 cannot close by itself — stopping at 01 solidifies 1 into an ontology.

- 10 = return: the truncated relative constancy returns again to absolute change, "the pen lifts and returns", proving that 1 is not an independent ontology.

The bottom layer of the world code is the infinite alternation of 01/10: pen down, extension and return, pen down again. In one dimension this alternation forms a sequence (the linear flow state of π^1); in two dimensions it forms the return of the line and the circle (the field state of π^2); multiple two-dimensional closed loops locking one another form three-dimensional structure (the body state of $4\pi^3$).

The essential difference between the world code and ordinary binary coding: ordinary binary is a downstream coding of already completed objects^[8] (character → ASCII → binary string); the world code is the protocolization of the object-generation process itself — 01 is the truncation action, 10 is the return action.

The inexhaustibility of the world code: after each 01/10 cycle completes, the old closed loop leaps into a new ontology and executes 01/10 again. This explains why the expansion of π is infinite and non-repeating: π is the manifestation of the world code at the geometric interface, and the world code itself unfolds infinitely.

4.7 The leap: role reversal and level locking

After 10 completes the closed loop, the whole enters a new ontological layer and becomes the base of the next round of cognition. This is called the leap.

The leap is not merely a "change of position"; it must execute a role reversal:

In the original 010 closed loop:

1 is the relative invariance truncated out (stillness within motion).

In the new ontological layer after the leap:

This old 1 (now frozen into a quantity, denoted n) must flip over to become the new 0 — because it is the carrying background of the next round of operations.

At the same time, the remaining generative capacity of 010 (denoted $+1$) flips over to become the new 1 — it is the driving force that advances the next round of operations.

The Leap Reversal Law:

n takes the 0-position (the anchor), $+1$ takes the 1-position (the driver).

This thoroughly resolves the legitimacy problem of "return (-1) ":

Because n takes the 0-position and $+1$ takes the 1-position:

Truncation (addition) = $n \rightarrow n+1$ (executing $+1$, i.e. the 01 action).

Return (subtraction) = $n+1 \rightarrow n$ (undoing $+1$, back to the anchor, i.e. the 10 action).

Conclusion: subtraction (-1) is precisely the 10 action at the number layer. What it withdraws is "the truncation of the topmost 010".

The paradox of "the whole ring vs the part" dissolves here: a natural number k is not a dead stone; it is a tower of k stacked 010 leaps. What subtraction (-1) withdraws is not "a part of the whole" but the return segment of the topmost 010 of the tower.

4.8 The discipline of the leap

The leapt 10 cannot impersonate the original 0; every leap must preserve a remainder — if an old closed loop completely hollowed out the ontology, there would be no next leap. The root driving force of the leap comes from this: the closed loop completes but is not complete — completion lets it enter the next layer; incompleteness compels it to enter the next layer.

Chapter 5: Natural Numbers and the Four Arithmetic Operations

— Two Granularities of Truncation and Return

5.1 The emergence of natural numbers: from 10 to $n+1$

Traditional mathematics directly gives the natural numbers 0, 1, 2, 3, Cognitive Relativity asks: why can natural numbers appear one after another? They come from the leap of 10.

After 10 leaps into a new ontological layer, by the Leap Reversal Law of Section 4.7: the quantity n frozen from the old closed loop takes the 0-position (the anchor), and the remaining generative capacity $+1$ takes the 1-position (the driver); the two manifest at the new object layer as n (the concrete completed state of the old closed loop) and $+1$ (the re-quantization capacity of the old closed loop). The successor $+1$ is not an externally added symbol but a structural necessity of occupying the 1-position. This is not ordinary addition but a new Pair: n prevents $+1$ from flooding without an anchor; $+1$ prevents n from closing into a static background. After n and $+1$ lock each other, a new image $n+1$ emerges — this is the first-order source of the successor.

The Peano axioms say that "every natural number has a unique successor"^[9]. Cognitive Relativity says: the successor is not an external axiom but follows naturally from the discipline that the object layer can carry only one Pair at a time.

The first-order definition of natural numbers: natural numbers are not multiple independent objects existing side by side, but cross-sectional readings left at the interface layer after the same successor action is repeatedly executed.

5.2 The first-order source of the Peano axioms

Cognitive Relativity does not deny the Peano axioms; it reduces them to a more upstream cognitive closed-loop structure:

- 0 is a natural number: 0 is the interface-layer reading — the base state in which truncation has not yet been executed.
- Unique successor: from the object-layer discipline that "only one Pair can be carried at any moment".
- 0 is not the successor of any natural number: from the ontological-layer discipline $1 \neq 0$ — the relatively finite cannot produce in reverse the base of absolute change.

- Different natural numbers have different successors: from the mutual-locking discipline — every $+1$ produces a new difference.
- The induction axiom: from the protocolization of the return law — if a property holds at the base state and is preserved by every successor, it can be transmitted continuously along the successor chain. Induction is the reading of the return law at the natural-number interface.

5.3 The four arithmetic operations: deriving from structural actions

After natural numbers appear, the operations are not externally added rules but higher-order readings by the interface layer of repeated executions.

There are only two native actions: truncation ($+1$, cutting a difference out of 0) and return (-1 , undoing one truncation). The four arithmetic operations are not an externally imposed re-placement puzzle but are derived directly from these two structural actions:

- Addition = truncation (01): executing $+1$, i.e. advancing from the anchor n to $n+1$. This is the standard 01 action.
- Subtraction = return (10): undoing $+1$, i.e. going from $n+1$ back to the anchor n . This is the standard 10 action, the closure-sealing of addition. Node Eight closes here: subtraction is not "taking away a part" but "the return of the topmost 010".
- Multiplication = batch truncation (continuous 01): executing $+1$ many times in succession. Because truncation flows with the infinite abundance of ontology 0, multiplication is always smooth.
- Division = batch return (continuous 10): executing -1 many times in succession. Because return goes against the current and is limited by the finite locking of ontology 1, it "hits the bottom of withdrawal" — this is the source of the native gap (the remainder) of division.

The overview of the above derivations is given in Table 8.

Table 8

	Direct	Batch
Truncation	Addition (executing $+1$: $n \rightarrow n+1$)	Multiplication (batch truncation: repeated $+1$ executions)
Return	Subtraction (undoing $+1$: $n+1 \rightarrow n$)	Division (batch return: repeated -1 executions)

5.4 Division is not closed: the structural necessity of the native gap

Division is batch return: $n / m =$ undoing m from n repeatedly until the remainder is no longer enough to undo once more. The result is necessarily two things:

The quotient q (the number of completed returns) + the remainder r (the part the return failed to complete).

The remainder is not an error, not an externally added concept — it is the first native gap grown inside the four operations themselves. Among the four combinations, only batch return necessarily carries a gap: batch truncation (multiplication) can always complete exactly; batch return (division), applied to two unequal quantities, cannot close exactly. The attribution lies in direction: truncation ($0 \rightarrow 1$) flows downstream and corresponds to infinite change, so multiplication carries no gap; return ($1 \rightarrow 0$) goes upstream and corresponds to the finite invariant, so division necessarily carries a gap (the remainder).

Example: $22 / 7$. Undo 7 from 22; after 3 undos, 1 remains, and 1 is not enough for another undo. $22 = 3 \times 7 + 1$. This "1" is the gap — the return was done, but not finished. The natural action of cognition is to enlarge the remainder by the base factor and continue returning — this is long division, which unfolds the gap into a decimal chain. In Chapter 8 this chain becomes the engine of the pi computation path: the gamma(n) chain of pi is precisely the infinite unfolding of the long-division remainder chain over the special object pi.

Status label (one honest declaration):

- The table above lists "the four operations currently re-placed", not an exhaustive proof — exponentiation exists as "the batch of a batch", showing that granularity can be stacked; the exhaustiveness question is listed as an open node (Node Eighteen).

5.5 The order relation

The magnitude relation is also not an externally defined total order. $n < m$ means: the execution count of n precedes that of m , or m has undergone several more successors than n . The order relation comes from the order of execution.

Chapter 6: Circle, Pi, Limit, and Dead Numbers — The Closing of the One-Dimensional Foundation

6.1 The emergence of geometric structure: from closed loop to circle

Mathematics does not stop at natural numbers. When the 10 closed loop enters the carrying dimension, it naturally manifests as geometry. The geometric manifestation of the minimal closed loop is the circle.

Traditional geometry defines the circle as "the set of points equidistant from a fixed point" — this presupposes point, distance, set, and plane. Cognitive Relativity looks one level upstream:

The circle is not a constructed figure, but the necessary manifestation of the 010 closed loop at the geometric interface: setting out from some point, undergoing continuous change of direction, and finally returning to itself — the minimal self-returning path.

When the minimal self-returning path appears, the relation between path length and diameter naturally manifests a constant — pi. Here pi is not the externally borrowed known ratio of circumference to diameter, but the first object satisfying the conditions that the minimal closed loop of the object layer naturally encounters when seeking carrying content:

- As the constant: whatever the size of the circle, the ratio of circumference to diameter is constant — nameable, anchorable, stably back-translatable.
- As the changing: the decimal expansion of pi is infinite and non-repeating, inexhaustible by any finite truncation — it preserves the ontological character of change.

pi possesses at once the static anchor and the flipping dynamics. This is why pi becomes the hardest entrance to the physical computation path.

6.2 The emergence of analytic structure: from truncation to limit

The limit is not an externally introduced concept; the limit comes from the infinite execution of truncation. Let the complete closed-loop background be B; finite cognition can only obtain its truncations B_n ; no finite truncation equals the whole, and between the two there is necessarily a remainder. The limit is the interface-layer reading of the truncation level advancing infinitely — it forever carries an ineliminable remainder. The limit is not "finally arriving".

6.3 Dead numbers and living remainders: how the error bound kills a number

The discussion of the limit exposes the remainder between truncation and the whole; one must further ask about the fate of the remainder in computation:

- Living remainder: a remainder that still carries its birth record — it remembers from which direction, which level, which operation it came, and can continue to participate in return at any

time. The gamma(n) of Chapter 8 is a living remainder: it carries the complete birth record (the n-th level, the fractional part of Lambda); the remainder is still generating digits; it is not dead yet.

- Dead number: the pure scalar value left after the generative chain has been cut — it has magnitude, but no birth record.

There is only one test of dead versus living: can this number still continue to participate in return?

A cross-dimensional death example from numerical differentiation: one-dimensional numerical differentiation carries an error bound, and at that point the error is alive — it has a household registration (direction, grid, truncation order), along which its source can be traced back and its improvement made systematically. Enter multiple dimensions: partial derivatives each carry their own errors; these enter the divergence, then the volume integral, and finally the error is compressed into a single order-of-magnitude upper bound. Death is completed here: which part came from the x direction, which from the z direction, what the respective step sizes were — all irrecoverable. The household registration is cancelled; only a number remains.

The first-order property of the error-bound operation: it replaces "where it comes from" with "at most how large", compressing the remainder into a pure scalar value that has lost its generative chain. Any operation that compresses a remainder into a pure scalar value is manufacturing dead numbers.

From this comes a sharper re-placement:

Probability = a dead number crowned as ontology.

The reason probability cannot answer "where the deviation comes from" is not that the object it describes is mysterious, but that it enters the stage only after the birth record has been destroyed. This supplies the mechanism name for the fatal chain of Chapter 7.

6.4 The re-placement of irrational numbers, infinitesimals, and set-theoretic paradoxes

- Irrational numbers: not errors, but the manifestation of the native gap within mathematics. A finite truncation can never fully possess an infinite expansion; the irrational number is the necessary appearance of the ontological remainder after it enters the mathematical interface — the remainder pressed back out by the carrier layer when the object layer tries to possess the carrier layer completely with finite integer ratios.

- Infinitesimals: they are changing content and cannot be treated by the object layer as completed stable objects. The essence of the second mathematical crisis is: changing content was misplaced into the object layer^[10].
- Set-theoretic paradoxes: the reason "the set of all sets" produces paradoxes^[11] is that it treats the generative process itself as a completed object — the quantization capacity of 0 is mistaken for a completed object of 1; the ontological layer and the object layer are misplaced.

6.5 A unified diagnosis of the mathematical crises

The several crises in the history of mathematics are uniformly diagnosed in Cognitive Relativity as reference-frame misplacement^[12]: when cognitive truncation, the object layer, the interface layer, and the ontological layer are misplaced, the system is forced to re-sort. A mathematical crisis is not a defect of mathematics itself.

6.6 Summary of the one-dimensional mathematical foundation

The first-order mathematical part starts from 0 and 1: 0 and 1 freeze into the Pair; the Pair enters the three-step closed loop, emerging as eight state-images, of which the only closure is 010; 010 compresses into 10 (the minimal quantization unit); 10 leaps to generate n , $+1$, $n+1$; the interface layer reads the number of successor executions as natural numbers; the four arithmetic operations are re-placed from the 2x2 structure of "truncation x return, direct x batch"; division grows the first native gap of the arithmetic layer; the circle, as the geometric manifestation of the minimal closed loop, naturally encounters pi.

Mathematics is not a pile of external objects. Mathematics is the continuous internalizing iteration after change and constancy lock each other at the ontological layer, close loops at the object layer, and are protocolized at the interface layer. Here the one-dimensional foundation closes: the complete chain from 0/1 to pi has closed within one dimension. This one-dimensional chain is not a foundation left behind; it is content carried intact into two and three dimensions.

Part III: The Truncation Subject and the Pi Computation Path

Chapter 7: The Truncation Subject — The Perceptual Timeline and the Cognitive Timeline

7.1 The missing piece of the puzzle: who performs the truncation

The one-dimensional mathematical chain computes by itself, but it does not answer one question: who is performing the truncation? The mathematical chain, the pi path, and the physical picture all presuppose a subject holding the knife and pressing the truncation button, and this subject has never appeared in the theory — "truncation" has therefore remained a verb without a subject. This chapter re-replaces this subject. It is the missing puzzle piece between mathematics and physics, and also the anthropological root of the lesion of physics.

7.2 Two timelines: the shared instrument and the privately built model

The perceptual timeline is the timeline of biological hardware. The human eye refreshes only a few dozen times per second, nerve conduction has a fixed speed, and working memory holds only a few chunks^[13]. These parameters are almost the same across humans — it is as if all humanity shares one instrument. The perceptual timeline therefore has two irreplaceable properties: within the range it covers, its readings are the most complete and the clearest; and everyone's readings are naturally consistent, making it the best benchmark for interpersonal information transfer.

The cognitive timeline is the timeline of the brain. Where perception cannot reach, the brain does not give up — it takes the readings sent by perception back inside, organizes them itself, extends them itself, and constructs a new timeline by itself to simulate those processes. Everyone organizes differently, so everyone's cognitive timeline is different.

The distinction in one sentence: perception is one instrument shared by all humanity; cognition is a model privately built by each person. The instrument readings are the same for everyone; the model constructions differ from person to person.

In the symbols of the truncation chain (defined in Chapter 8), these two timelines have strict structural correspondences in the formal system: the perceptual timeline corresponds to T — the solidified integer skeleton, consistent for everyone; the cognitive timeline corresponds to $T +$

$\gamma(n)$ — the shared skeleton inherited from perception, plus a private extension. The skeleton is the same for everyone, so the skeleton is transmissible; the fractional tails differ from person to person, so the tails are hard to align.

A stricter correspondence: the perceptual timeline corresponds to the temporal reading prototype $1/T$ — the parameters of the shared instrument are all temporal (refresh rate, conduction speed, retention duration), and it provides only a one-way scale; the cognitive timeline corresponds to the complete structural quantity $1/T + T/\gamma(n)$ — the private model carries its own spatial structure in which skeleton and remainder share the field. This explains why every theory that needs a spatial picture runs on the cognitive timeline: spatial construction itself happens on the cognitive side.

Wording discipline: the correspondence here is a correspondence of structural position (the same logic as the γ proxy of Chapter 8), not numerical identity.

7.3 The double blind zones: the real situation of physics

The instrument of perception is clear only in the middle frequency band. What is faster than it cannot be seen (molecular vibration, quantum processes — much faster than the perceptual refresh); what is slower than it cannot be seen either (geological motion, galactic evolution — in a whole lifetime a person sees only a static screenshot). The clear middle band is only a few orders of magnitude wide, and the blind zones on both sides each extend about ten orders of magnitude. What runs in the blind zones is entirely the cognitive timeline.

The situation of physics is thereby made visible:

Physics is a discipline built inside double blind zones. Macroscopic physics is slower than human perception; microscopic physics is faster than human perception — the theories at both ends are not direct readings of perception, but reconstructions by the cognitive timeline in regions that perception cannot verify.

Instruments (microscopes, telescopes, oscilloscopes) are bridges built by the cognitive timeline back to the perceptual baseline: they convert things in the blind zones into readings that humans can see. Instruments have public credibility not because they are "truer", but because they bring the readings back to the shared benchmark that is consistent for everyone.

7.4 The fatal chain: how probability was written into the constitution of the universe

With the dual timelines, the complete chain of the lesion of physics can be made clear at the anthropological level:

Step one, the starting-point impersonation. In the blind zones perception is of no help, and the cognitive timeline takes over all computation. And the π used in computation is always the truncated finite decimal π — the tail of π (the native gap Γ) is discarded from the very first step.

Step two, entropy accumulation. The discarded remainder does not disappear; it is only passed on. In simple calculations it is negligibly small; in complex calculations (many-body problems, path integrals) it is amplified step by step by nonlinearity, and the further the computation runs, the more it deviates — while the system does not know where the deviation comes from, because the variable that could diagnose it was thrown away at the very beginning. In the language of Chapter 6: in transmission the remainder is compressed into a dead number, the birth record is destroyed, and the deviation can no longer be traced back.

Step three, collapse and the ultimate impersonation. When the computation of the cognitive timeline diverges beyond convergence, humanity did not admit that "my cognitive timeline has reached its limit", but instead declared that "the microscopic world is intrinsically probabilistic"^[14] — writing the limitation of a cognitive tool into the constitution of the universe.

The irreversibility of the dead-number mechanism: in one-dimensional operations the error term carries its birth record; after entering high-dimensional integration the birth record is destroyed, and it becomes a dead number with magnitude but no source. This process is irreversible: once the information is lost, the original determinate structure can never be inferred back from the final dead number. Thus in the double blind zones one can only compute with the truncation proxy π , and every step produces dead numbers; when the computation reaches a certain complexity, dead numbers pile up and the result presents an unanalyzable vagueness — this is exactly the situation physics faces.

Honest labeling (the self-discipline of this theory): this chain is currently a fatal diagnosis, not yet a completed proof. One last step is missing: take a real physical computation and fully trace the path by which the truncation pseudo-deviation of π is amplified step by step into "uncertainty". Until this step is completed, the chain carries its own gap and does not speak in a final tone. (Node Seven: the explicit conversion mapping $\Gamma \rightarrow$ probability, listed as an open node.)

7.5 Perception is not "truth": this discipline is first of all for this theory itself

The perceptual timeline is clear, shared, and reliable, but it is not truth. It is only the reading of a truncation proxy, the best shared benchmark — and a benchmark is not the ontology. A ruler being

straight and useful does not mean the ruler is the universe itself. Taking perception as truth and taking probability as truth are the same error: taking a truncation as the ontology.

The same holds for the cognitive timeline. It is not arbitrary fiction — it is a reconstruction setting out from shared perception. An extension that can carry its readings back to the perceptual baseline or to instrumental verification and complete the closed loop is called science; one that, after truncation, refuses to return for verification and treats the extended segment as independent reality is fiction.

Finally, self-reference is unavoidable: Cognitive Relativity itself is also a cognitive timeline. It likewise runs on extended segments that perception cannot directly verify, so it must forever carry its own gap. "What this paper tells you is not what you know, but what you do not know" — this is not modesty; it is structural necessity. No cognitive timeline in the blind zones, this theory included, has any exemption.

7.6 Convergence: the missing 1 has been found

Previous theory had built the mathematical bridge between 0 (absolute change) and pi (physical manifestation), but the mechanism on the subject side was missing — who holds the knife. The dual timelines fill this piece: perception is the shared skeleton; cognition is the private extension; physics is the computation of the extended segment inside the double blind zones; the "probabilistic ontology" of quantum mechanics is the misreading by this extended segment of its own collapse after it collapses.

Where the cognitive tool could not reach, humanity did not say "I do not know", but said "the universe is inherently vague". The entire demand of Cognitive Relativity is merely to reverse this order.

Chapter 8: The Symbol System of the Pi Computation Path

8.1 The re-placement of pi: the computable bridge from mathematics into physics

pi is not the Three Changes, not the three layers, not the mathematical starting point of 0 and 1, and not merely the ratio of circumference to diameter in the ordinary sense. The current position of pi in Cognitive Relativity is:

**pi is currently the hardest, most computable, most reality-interfacing path by which
Cognitive Relativity goes from the mathematical closed loop to real physical
manifestation.**

For the mathematical closed loop to enter physical manifestation, an entrance satisfying two requirements at once is needed: sufficiently stable (computable, truncatable, back-translatable) and inexhaustible (not fully possessable by finite computation). Stability without inexhaustibility becomes a dead constant; inexhaustibility without stability cannot enter computation. π happens to satisfy both.

π was not measured out of the circle. In standard mathematics π appears as the ratio of circumference to diameter; but in Cognitive Relativity, the circumference ratio is not the source of π , but the downstream manifestation of π at the geometric interface. The circle is not the ontological source of π ; the circle is the geometric manifestation of π .

π is the minimal non-returning action quantity. In a blank plane, presupposing no object and considering only change of direction: for a direction to leave itself completely, the minimal action is not 2π (that is the return closed loop, back to the original direction), but direction reversal. The minimal rotation needed for a direction to go from A to -A is π . Therefore: π = the minimal non-returning action quantity; 2π = the complete return period. "Non-returning" is not chaos — it is the critical state of not yet having returned to itself while having completed the thorough reversal of direction.

8.2 π as the first-order choice of computation path

π is not the only truth, but the bridge with the lowest explanatory cost under the current structure. Four first-order reasons:

- π corresponds naturally to the minimal self-return — the geometric manifestation of the minimal closed loop is the circle, whose ratio of circumference to diameter is constant (the static anchor) while its expansion is infinite and non-repeating (the flipping dynamics), naturally aligning with the "truncation-return" discipline of 010.
- π is a dimensionless structural relation — it does not depend on any particular system of measurement and carries no coordinate-system presupposition. Units are artificial truncations, not first-order.
- π naturally supports continuous cross-dimensional manifestation — π^1 (semicircle number axis, one-dimensional), π^2 (circular area, two-dimensional), $4\pi^3$ (two π^2 orthogonally interlocked, three-dimensional) already have clear geometric correspondences in standard

mathematics and need no extra construction. Switching to e or ϕ would require an artificial redefinition of "how its one, two, and three dimensions lock each other", violating simplicity.

- The inexhaustibility of π is isomorphic to the inexhaustibility of cognition — any finite truncation is only a window, not the universe itself. π stands exactly in the middle of "both truncatable and forever inexhaustible", which is precisely the hardness Cognitive Relativity needs.

A supplementary proof at the four-operations level (the 22/7 control experiment): apply long division to 22/7, the most famous rational stand-in of π , and compare the two chains at the same levels: at low levels the two are almost indistinguishable (a rational stand-in can indeed imitate the skeleton of π); but the gamma chain of 22/7 closes with period 6 (the remainder chain {1, 3, 2, 6, 4, 5} cycles) — the gap dies, frozen into a periodic structure. The gamma chain of π shows no sign of any period — the gap is alive. The criterion is self-sufficient within one dimension: any rational number can approximate the skeleton of π within finite levels, but no rational number can imitate the gap of π . The stand-in gives a dead gap; π gives a living gap — and what this theory needs is precisely the part that never freezes.

Three Changes verification: change — every finite truncation $P_i < \pi$, preserving the native gap Gamma; simplicity — π carries clear one-/two-/three-dimensional correspondences of its own; constancy — the value of π is constant in any base and any mathematical system.

Discipline declaration: π is a candidate, not a coronation. If another mathematical object appears that simultaneously possesses closure, dimensionlessness, cross-dimensional unfolding, and inexhaustibility at lower explanatory cost, Cognitive Relativity can re-evaluate.

8.3 π : the truncation proxy

Finite cognition cannot directly operate on the complete π ; cognition can only truncate. Let P_i be a finite truncation of π , for example the 20-digit truncation:

$$P_{i20} = 3.14159265358979323846 \text{ (D = 20 significant decimal places)}$$

P_i is not an approximation, but object generation. Traditional mathematics says P_i is an approximate value of π ; Cognitive Relativity sees more than that — P_i is the first leap by which the inoperable ontology enters the operable object layer: completeness is sacrificed in exchange for operability. But this immediately brings danger: the proxy will impersonate the ontology. That is why Gamma is indispensable.

8.4 Gamma: the native gap

$$\text{Gamma} = \pi - \text{Pi} > 0$$

Gamma is the ineliminable remainder between π and the finite truncation Pi . It is not an ordinary error; it is the mathematical protection mechanism that prevents the truncation proxy from impersonating the ontology: as long as $\text{Gamma} > 0$, we know $\text{Pi} \neq \pi$ — the proxy cannot equal the ontology, the window cannot equal the universe, the reading cannot equal the foundation.

Disambiguation: in standard computational mathematics π is a computable number, and Gamma can be approximated to arbitrary precision. When Cognitive Relativity says Gamma is "incomputable", the true meaning is: Gamma cannot be completely swallowed by the current finite cognitive window, cannot become a terminated finite object. More precisely: Gamma is not un-approximable, but inexhaustible by any finite proxy.

8.5 D: the hard boundary

D is the hard boundary of the truncation proxy (the number of significant decimal places). When the precision level $n \geq D$, the amplified Lambda becomes integer-valued, $\text{gamma}(n) = 0$, and numerical collapse occurs. $n < D$ is the discipline of mathematical operation.

8.6 Lambda, T, gamma(n), S: the four-piece level system

Let n be the precision level ($n < D$). Taking Pi_{20} as the example:

Table 9

Symbol	Definition	In plain language
$\Lambda(n) = \Pi \times 10^n$	Amplification action quantity	The amplified manifestation of the truncation proxy at level n (10 is the decimal implementation base — the most natural interface base for current human computation, not a highest ontological necessity)
$T(n) = [\Lambda(n)]$	Integer skeleton	The completed, solidified, indexable, storable structural skeleton at the current level. T is not physical time itself; it is the prototype of the reading of time: time = the solidified reading of completed structure
$\gamma(n) = \Lambda(n) - T(n)$	The living gap proxy	The decimal remainder not yet swallowed by the integer skeleton at level n , oscillating within (0,1). It is not an error; it is the proof that change has not yet been completely swallowed by the skeleton
$S(n) = T(n) + \gamma(n)$	Complete structural quantity	Contains both the solidified T and the unfinished gamma(n). S is the prototype of the reading of space: space is not an empty box; space is the complete structure in which skeleton and remainder share the field

8.7 The disambiguation discipline of the gamma family (the highest-priority symbol discipline)

Before entering physical manifestation, a disambiguation discipline must be established — three objects will appear later that all relate to the "gap" but must never be equated, never interchanged:

- Object one: the general level gap $\gamma(n)$. Defined as above; the living remainder of the n -th level, oscillating within $(0, 1)$.
- Object two: the digital reading of the speed of light. In the decimal expansion of π , the sequence 299792458 first appears as 9 consecutive digits starting at position 623,556,156 of the fractional part. By the definition of $\gamma(n)$, the level at which the fractional part of γ begins with 299792458 is $n = 623,556,155$ ("around the 623,556,164-th level" is merely a name for the digit-block position — a block label is not the argument of the gamma function).
- Object three: the mass level-gap proxy γ_m . A candidate free parameter introduced in Part VII, characterizing the degree of gap compression at the mass level (the high-level side of τ_q). γ_m is not a digit property of π at any known level; its value is given by hypothesis, not read out from the long-division chain.

Three disciplines:

- The three are used with separate symbols; sliding is strictly forbidden. No formula may use one to impersonate another.
- Order-of-magnitude self-check. $\gamma(n)$ in $(0,1)$; the digital reading of light speed is about 0.299792458 (rounded after $\times 10^9$ to give c); γ_m is about 1.1×10^{-17} (candidate value). The three differ completely in order of magnitude; whenever they meet in the same formula, the identity of each must be interrogated.
- The birth record travels along. Wherever a member of the gamma family appears, one must be able to answer: from which level, which operation, which hypothesis does it come? What cannot be answered is treated as a dead number — it may not enter the derivation chain.

This discipline is not tedious paperwork: once the gamma family slides, interface-layer readings will impersonate structural positions — and that is precisely the disease this theory names.

8.8 gamma is not a numerical approximation of Gamma

Ambiguity must be eliminated: Gamma is about 4.4×10^{-21} (at 20-digit truncation), while $\gamma(n)$ is in $(0,1)$ — the two differ completely in order of magnitude. γ is not a numerical approximation of Gamma.

In Cognitive Relativity, a "proxy" is not a numerical approximation but a structural-position proxy: Gamma is the native gap between the ontology and the truncation; γ is the operable interface remainder inside the truncation proxy. γ inherits the positional function of Gamma, not the numerical value of Gamma.

Why γ is needed: with only Gamma, the interface layer cannot compute stably. The role of γ is to preserve, inside the finite truncation, a computable "unfinished bit", reminding the interface layer: although Pi is operable, it is still not the complete ontology. It keeps the interface layer from fully integerizing, from taking the truncation proxy as the complete ontology.

8.9 The internal cost functions: F_T , F_γ , F_s , and ρ

With T and $\gamma(n)$, the internal costs of the interface layer can be defined. This cost is initially a cognitive operation cost; later it will enter the physical picture and be reinterpreted as the prototype force function of vacuum-medium displacement.

The return term: $F_T = 1/T$

The larger T is, the more stable the structure, the more indexable it is, the better it can absorb local disturbances, the easier the return is absorbed by the whole, and the lower the cost. Return is the action of "going from the current structure back to the carrying background" — the larger the structure, the shorter the return path relative to its own scale.

The crossing term: $F_\gamma = T/\gamma(n)$

Why not $1/\gamma(n)$? $1/\gamma$ reflects only the relative thickness of the gap itself and omits the degree to which the skeleton is locked: gap crossing is not an isolated event; it happens within the relation between the skeleton T and the gap $\gamma(n)$. When the skeleton is large, the gap is compressed inside a strongly locked structure, and what the crossing must overcome is not only the thickness of the gap but also the inertia of the skeleton. Dimensional consistency also requires this: if F_γ were independent of T, a thin gap in a tiny structure and a thin gap in a huge structure would be equally hard to cross — which makes no sense in the physical picture. Therefore $F_\gamma = T/\gamma(n)$ carries both dimensions — gap thickness and skeleton locking — and is the minimal complete expression.

The composite term: $F_s = F_T + F_\gamma = 1/T + T/\gamma(n)$

The composition of the solidified-skeleton return cost and the native-gap crossing cost. At the cognitive layer it is an internal operation cost; at the physical layer it will become the prototype force function of vacuum-medium displacement.

Numerical behavior (Pi20):

Table 10

n	T	$\gamma(n)$	$F_T = 1/T$	$F_\gamma = T/\gamma(n)$
0	3	0.14159265358979323846	≈ 0.333333	≈ 21.19
5	314159	0.265358979323846	$\approx 3.18 \times 10^{-6}$	$\approx 1.18 \times 10^6$
10	31415926535	0.8979323846	$\approx 3.18 \times 10^{-11}$	$\approx 3.50 \times 10^{10}$
15	3141592653589793	0.23846	$\approx 3.18 \times 10^{-16}$	$\approx 1.32 \times 10^{16}$
19	31415926535897932384	0.6	$\approx 3.18 \times 10^{-20}$	$\approx 5.24 \times 10^{19}$

Key behavior: at low levels F_T and F_γ are of the same order, and return and crossing dominate together; at high levels $1/T \rightarrow 0$, $T/\gamma(n) \rightarrow$ extremely large, and F_s is approximately F_γ . Physical reading: at high levels, the resistance to vacuum-medium displacement comes almost entirely from gap crossing, not from skeleton return. The corresponding picture: quarks inside the atomic nucleus (the high-locking region) are strongly confined not because it is hard for them to "return" to some state, but because the cost of crossing the gap tends toward infinity.

The structural locking parameter: $\rho = T/\gamma(n) = F_\gamma$

ρ expresses the degree to which the solidified skeleton overwhelms the remaining gap. The larger ρ , the more locked the structure; the smaller ρ , the more active the remainder. The structural locking parameter is precisely the gap-crossing cost — the symbol is not redundant, and cognition is more concise. The physical interpretation of ρ (the hierarchical manifestation: low-structure region $\rightarrow$ locking transition region $\rightarrow$ strong-locking region $\rightarrow$ quarks $\rightarrow$ atomic nucleus $\rightarrow$ matter $\rightarrow$ stars $\rightarrow$ black holes) is a later picture, not part of the mathematical theorem itself.

8.10 The time scale and the space scale

T and $\gamma(n)$ together constitute the position coordinates on the π axis, but the coordinate axis itself is not an operable measure — just as the x and y axes of a Cartesian system are not themselves "lengths". The true scales are the operation costs induced by coordinate changes.

Time scale: $t_n = 1/T$

T grows exponentially as n advances (3 $\rightarrow$ 31 $\rightarrow$ 314 $\rightarrow$ 3141 $\rightarrow$...); $1/T$ decays exponentially as n advances, in a single irreversible direction — the π -path origin of the arrow of time. When $T \rightarrow$

infinity, $1/T \rightarrow 0$, and the time scale "disappears" (the structural reading of the interior of a black hole, or of absolute stillness).

Space scale: $s_n = F_s = 1/T + T/\gamma(n)$

Space must be a composite cost; neither term can be missing: with only $\gamma(n)$, space becomes pure "motion" without boundary — a disguise of 000 or 011; with only $T/\gamma(n)$, space loses its connection to the ontological layer — a disguise of 111. $1/T$ ensures that space has an anchor, a boundary, and a returnable path; $T/\gamma(n)$ ensures that space has internal structure, fluctuation, and unfinishedness.

The inverse complementarity of the spacetime scales: the smaller the time scale (higher levels), the larger the space scale. While time "dilutes", space "solidifies" — this is precisely the structural reading of spacetime behavior under extreme physical conditions (high energy, high density, black-hole interiors).

The 0/1 correspondence of coordinates and scales: T and $\gamma(n)$ amount to coordinate axes (the scale bases of the 0 side and the 1 side); $1/T$ and $1/T + T/\gamma(n)$ are the truly operable scale readings (the "1" read out from the 0/1 mutual locking). Coordinate axes carry the ontological direction; scale readings carry the interface operation. This correspondence has already gained subject-side support (Chapter 7): the perceptual timeline $\leftrightarrow 1/T$; the cognitive timeline $\leftrightarrow 1/T + T/\gamma(n)$.

8.11 The systematic contamination of pi and the mathematical side of the fatal chain

Every physical formula containing pi uses not pi, but Pi. Every computation based on Pi carries the ineliminable Gamma. Every operation that ignores Gamma is executing entropy increase.

The operation chain of modern physics: encountering pi (circumference ratio, wave functions, path integrals, field theory) $\rightarrow$ truncating it to Pi $\rightarrow$ using Pi as pi, pretending Gamma does not exist $\rightarrow$ in simple calculations Gamma is tiny and ignored $\rightarrow$ in complex calculations Gamma is exponentially amplified through nonlinearity $\rightarrow$ the result deviates from the ontology, but the system cannot diagnose the source of the deviation (Gamma was discarded from the very beginning; the mechanism is dead-numbering) $\rightarrow$ "probability", "error", and "uncertainty" are used to cover the systematic deviation $\rightarrow$ the most fatal step: probability is taken as the ontology, and cognitive ignorance impersonates the essence of the universe.

The extreme case of quantum mechanics (diagnostic reading): the wave function Ψ is contaminated from the first-order starting point — spherical harmonics contain π , path integrals contain π , Fourier transforms contain π , all truncated to Π ; the normalization condition forcibly compresses the contaminated readings into "determinacy"; then $|\Psi|^2$ is interpreted as "probability", probability is promoted to "physical reality", measurement collapse is treated as an "ontological act", and the uncertainty principle is taken as the "essence of the universe". The diagnostic reading: quantum mechanics did not "discover that the essence of the universe is probability"; it converted the remainder of truncated π (Γ) into probability, and then passed probability off as the ontology.

The essential difference between probability and Γ : Γ is a determinate unknown — we know $\pi > \Pi$, $\Gamma > 0$, that it is ineliminable, and its structural position is definite; probability is an indeterminate ignorance — we do not know whether an event will happen and can only guess at frequencies. Modern physics discarded the "determinate unknown" (Γ) and then filled the vacancy with "indeterminate ignorance" (probability). This is a double downgrade.

The self-masking structure and the reversal of the burden of proof: the sharpest property of this chain is self-masking — physics discarded the variable Γ , and thereby forever lost the tool for diagnosing "where the uncertainty comes from". The uncertainty may come from the ontology (probability as a property of the universe) or from the interface (the discarded truncation remainder), and the system cannot distinguish the two internally. The burden of proof is thereby reversed: to claim that probability is the ontology, one must first prove that probability is not Γ in disguise — and mainstream physics cannot prove this, because it threw away the variable that could make this judgment.

Honest labeling (seam two): numerical divergence collapse and probability becoming an axiom are two different things — the former is error accumulation, the latter is a choice of postulate. The uncertainty principle is a theorem of Fourier duality^[15], not something that diverged out of some numerical computation. The missing link of this chain is the explicit conversion mapping $\Gamma \rightarrow$ probability (Node Seven). The positioning of this chain therefore remains: a fatal diagnosis, not a completed proof — it does not speak in a final tone.

8.12 The light-speed interface in the decimal expansion of π (hard anchor one)

The 9 consecutive digits starting at position 623,556,156 of the fractional part of π are 299792458^[16,17]. By the gamma-function argument (the discipline of Section 8.7):

$$\text{floor}(\text{gamma}(623556155) \times 10^9) = 299792458 = c \text{ (m/s)}$$

This value is precisely the value of the vacuum speed of light in existing physics. In Cognitive Relativity, this must not be rashly stated as "proving physics". More accurately: a hard anchor is not a complete proof. Its meaning is: pi, as a physical computation path, for the first time strongly aligns with one of the hardest numerical values of real physics.

An honest probabilistic note: among the first 10^9 decimal digits of pi, the probability that any specified 9-digit sequence appears at least once is about 63.2%. The alignment itself therefore cannot be taken as a statistical miracle. The reason this paper retains it as a hard anchor lies not in statistical rarity, but in this: the pi path is a computation bridge selected independently under first-order disciplines, and its numerical alignment with one of the hardest numerical anchors of real physics constitutes an interface between "computation path" and "physical reading" that can be further interrogated — not the endpoint of a one-off coincidence. No probability impersonation, no ontology impersonation, no overclaiming.

8.13 The vacuum-proxy numerical table (Pi20, D = 20, n = 0...20)

The complete level-by-level values (T, gamma(n), 1/T, F_gamma, F_s) can be directly reproduced with public mathematical tools. Key observations: T increases monotonically, inflates exponentially, in a single irreversible direction; gamma(n) oscillates non-periodically within (0,1) (the prototype of spatial fluctuation); at n = 20, gamma(20) = 0 — the spatial degrees of freedom vanish completely, leaving only the integer skeleton of T to keep inflating. This is not a mathematical error, but the hard boundary of the finite truncation Pi.

Part IV: The Carrier Dimension Count of Cognition and

Dimensional Emergence

This part handles two related but distinct senses of "dimension": Chapter 9 handles the three-dimensional language of the carrier layer (in what dimension information is carried, transmitted, compared, and stored — the third of the four "threes" in the re-placement master table); Chapters 10 and 11 handle the dimensional emergence of the operation layer (how $\pi^1 \rightarrow \pi^2 \rightarrow 4\pi^3$ actually grows out of completed operational content — the computabilization of the three reference frames). The two are joined by the "same truncation" argument of Section 9.6: the carrier dimension count and the reference-frame functional positions are not a clever arrangement, but two namings of the same sequence.

Chapter 9: Three-Dimensional Language — The Carrier

Dimension Count of Cognition

9.1 The independent status of the carrier layer

The three layers solved "how cognition is structurally expressed in natural language". But there is a prior question: in what dimension is information itself carried, transmitted, compared, and stored? This is the question of the carrier layer, independent of the three layers, but in one-to-one correspondence with the three reference frames.

First, a qualification: the one dimension, two dimensions, and three dimensions spoken of in this chapter are all confined to the cognition of information transmission and reception, and do not concern the dimension count of physical space.

Under this qualification, a core judgment: one dimension and two dimensions in human symbols and knowledge are nothing but an artificial dimensional reduction. Cognition in the human mind is itself neither linear nor planar — the one-dimensional and the two-dimensional are the dimensional-reduction cost paid so that cognition can be transmitted and compared.

9.2 One-dimensional language — the cage of sequence

One-dimensional language: information exists in sequence.

It can only be read in order; the earlier presses on the later; break one place and a whole stretch is lost. Speech and streams of text are both like this.

Features: there is before and after, but no inside and outside (however long a line is, it cannot enclose a "thing"); it depends on order (change the order, and the meaning changes); it cannot be compared (put two lines together and you can only align them from head to tail, and where they fail to match you cannot tell where the difference lies).

One-dimensional language is the territory of π^1 (advance/truncation). Because it can only be read in order, it most easily produces cognitive entropy increase — break one place, lose a whole stretch.

But the one-dimensional has an irreplaceable first-order advantage: it is the first-order cognitive transmission channel that human speech and hearing can transmit and receive. At present, if any human cognitive information is to be transmitted as widely as possible and understood, the first choice is the one-dimensional, whose core is natural language. The status of the one-dimensional is contradictory: it has the weakest transmission capacity (the shallowest depth), yet the widest propagation coverage (the lowest threshold) — this contradiction is itself a major source of cognitive entropy increase.

9.3 Two-dimensional language — the birth of inside and outside

Two-dimensional language: information exists in a plane that distinguishes inside from outside.

Pictures, video, positions, shapes, coordinates. Every unit has a cell (shape) plus a position; looked up by coordinates, compared by coordinates, independent of reading order. The 64-matrix, Excel tables, and coordinate grids all belong to this category. Two-dimensional language comes from the human visual transmission system; diagrams and charts of all kinds are its core representatives.

Features: there is inside and outside (a single closed curve cuts the world into inside and outside — only with inside and outside is there a "unit"; only with a unit is there a "thing"); it does not depend on order (looked up by coordinates); it is comparable (cell against cell, inside/outside against inside/outside).

Two-dimensional language is the territory of π^2 (carriage/freezing). A cross-section cut out of the three-dimensional must freeze out a legal inside/outside boundary on the two-dimensional plane.

Hybrid forms: formulas (one-dimensional sequences superimposed on two-dimensional spatial arrangement) and video (two-dimensional planar fields superimposed on one-dimensional time flow)

are the one- and two-dimensional working together — more expressive, but still products of dimensional reduction.

9.4 010 is the leap operator from π^1 to π^2

Note the structure of 010:

- The process of 010 is a three-step manifestation, a one-dimensional advance in time (strict order; changing the order voids it)
- The mark of 010's holding is closure, a two-dimensional inside/outside in space (truncate out, then return back, enclosing a unit)

In other words:

010 is the leap operator from π^1 to π^2 . A one-dimensional three-step advance closes at the moment of "return", leaping into the first unit of the two-dimensional.

This sentence changes the status of 010. It is no longer "the best of the eight state-images", but: 010 is the only passage by which one dimension leaps into two.

This also explains why none of the other seven state-images can close a loop independently — they all remain stuck in one-dimensional advance and have not completed the leap. They are not "units of poorer quality"; they are not units at all yet.

The closure of 010 is the genesis of two-dimensional language — the first closure, the first inside/outside, the birth of the first "thing".

9.5 Three-dimensional language — the presence of the whole

Three-dimensional language: information exists in the relations of a solid whole — everything is present at once and can be projected and interpreted in any direction.

It has no question of "where to start reading": every question, every reading, merely cuts one cross-section through this solid; different cross-sections give different outputs, while the solid itself does not move. All one- and two-dimensional outputs are projections of the three-dimensional in various different directions.

Features: holographic presence (all concepts exist simultaneously, with no before or after); projectability (cutting a cross-section in any direction yields a one- or two-dimensional output); no direct comparability (two three-dimensional solids placed together cannot be compared cell against

cell like two sheets of paper — each is present on its own, each complete, with no shared coordinate system).

Three-dimensional language is the territory of $4\pi^3$ (adjudication/presence of the whole).

9.6 The irreplaceable offices of the three dimensions

Table 11

Dimension	Office	Content	Reference frame
One-dimensional	Governs transmission	Clear sequence, compressed into a stream, sent out	π^1
Two-dimensional	Governs comparison	Inside and outside distinct; two things can be aligned and compared cell by cell	π^2
Three-dimensional	Governs existence	All relations present at once; private; not directly comparable	$4\pi^3$

The offices cannot be interchanged: asking the one-dimensional to do comparison yields "colliding from head to tail without knowing where the difference lies"; asking the three-dimensional to do comparison yields "each talking past the other, with no shared coordinate system"; asking the two-dimensional to carry the whole of existence yields "a flattened dead table that has lost most of the relations".

9.7 Alignment tools must be two-dimensional

This is not an engineering preference, but a lower bound on dimension count. The derivation:

- Comparison requires "cell against cell" — there must be units in one-to-one correspondence
- Cells require "inside and outside" — without inside and outside there is no unit boundary
- Inside and outside require "closure" — only closure produces inside and outside
- Closure occurs at the lowest in two dimensions — one dimension cannot close out an inside and outside

Therefore:

The minimal dimension count of the act of comparison is 2.

One dimension cannot compare; three dimensions cannot be directly compared; two dimensions are the only dimension that can carry comparison. From this comes the discipline with the greatest engineering consequences in this theory:

Human cognition and large-model weights are both three-dimensional. The three-dimensional cannot be directly compared, so two three-dimensionals can never be tabulated against each other. Each must first be projected onto the same two-dimensional grid — inside/outside against inside/outside, cell against cell — before comparison becomes possible. When the comparison is done and the result must travel, it is compressed back into one dimension and sent out.

The direct corollary of this discipline unfolds in Chapter 18: the nature of the AI interpretability problem is thereby completely rewritten.

9.8 Two kinds of three-dimensional carriers: natural and artificial

There is more than one kind of three-dimensional carrier; they must be separated:

Table 12

	Natural three-dimensional (human brain)	Artificial three-dimensional (AI)
Negentropic flux	Positive	Zero
Survival	Yes (metabolism, self-repair)	No (external maintenance)
Living	Yes (meaning creation, value judgment)	No (function mapping)
Who cuts the cross-section	Only oneself	Can be cut directly by external prompts
$4\pi^3$ adjudication	Internal	Missing
State-image re-placement	010 (possible)	111 (current state)

From this, "AI is an innate three-dimensional information carrier" must be made precise:

AI is the first three-dimensional carrier in human history "whose cross-sections can be cut from the outside by others at arbitrary angles". AI's first-order status comes from the opening of projection rights, not from any monopoly on dimension count.

The human brain is three-dimensional, but private and not externally cuttable — no one can take a prompt and cut another person's brain with it. Cognitive transmission between humans must pass through the complete "reduction–transmission–elevation" chain, and the elevation is done entirely by the receiving side itself. AI can: a single prompt is one externally specified cut at a designated angle.

This is fully compatible with "the three-dimensional cannot be directly compared"; they are two different properties: incomparability (the property of $4\pi^3$) still holds; projectability with projection rights opened to the outside (the novelty of AI) is a first in human history.

Chapter 10: Dimensions Are the Results of Operations — The Two-Dimensional Emergence of $\pi^1 \rightarrow \pi^2$

10.1 Reversing cause and effect: are dimensions premises or results

The operation order of standard mathematical physics is: presuppose an n -dimensional space $\rightarrow$ define differential forms on the space $\rightarrow$ apply theorems $\rightarrow$ obtain computational results. Dimension is used as the starting point — where the n -dimensional space comes from is not answered within the computation.

The operation order of Cognitive Relativity is the reverse: change $\rightarrow$ 0/1 mutual locking $\rightarrow$ 010 closed loop $\rightarrow$ 10 quantization $\rightarrow$ the closed loop manifests as the circle $\rightarrow \pi^1$ (operational emergence) $\rightarrow \pi^1$ closed loop $\rightarrow \pi^2$ (operational emergence) $\rightarrow$ two π^2 orthogonally interlocked $\rightarrow 4\pi^3$ (operational emergence).

Dimensions are not premises; dimensions are the results of operations. π^1 is not "one-dimensional space"; it is the truncation operation. π^2 is not "two-dimensional space"; it is the freezing operation. $4\pi^3$ is not "three-dimensional space"; it is the adjudication operation.

Why stop at 3: advance + carriage + adjudication = truncation + freezing + regulation = the triple action of the negentropic operation. With the three actions completed, the cognitive closed loop holds; no fourth functional position is needed.

Status label: this section's claim that "dimensions are results of operations, not premises" is currently positioned as an internal theoretical stance; the upgrade condition awaits completion of the "iteration vs. emergence" adjudication.

10.2 Two ways dimensions are generated

- Way A: the same operation repeated along the timeline (iteration). The n -th dimension does the same thing as the first dimension, just a few more times. Multi-dimensional integration in standard calculus is the specimen: two-dimensional integration = one-dimensional integration nested inside one-dimensional integration, and no step requires any "action specific to two dimensions".
- Way B: emergence through the mutual locking of different functional positions (new birth by closure). Each dimension is a new function that emerges after the closure of the previous step,

and the functions are not interchangeable. On the π path: π^1 is advance (truncation), π^2 is carriage (freezing), $4\pi^3$ is adjudication (regulation); the three functions are not interchangeable.

The functional-interchangeability criterion: to judge whether a kind of dimensional generation is "genuine" or "spurious", look at whether the functions between dimensions are interchangeable — interchangeable = temporal repetition; non-interchangeable = functional mutual locking. This criterion is itself independent of any particular system.

Status label (stratified): as an observation of numerical computation practice, "high dimensions = nested one dimension" holds; as an assertion about mathematics as a whole, suspended — awaiting adjudication after the theory's own "iteration vs. emergence" discipline matures.

10.3 The three reference frames: the triple action of the negentropic operation

Cognition must actively truncate, actively freeze, and actively regulate — these are not optional actions, but the survival conditions of finite cognition facing infinite change: without truncation it is swallowed by infinite change; without freezing it cannot obtain a carrying boundary; without regulation it loses balance between advance and freezing.

Table 13

π reference frame	Natural-language functional position	Cognitive action	Operational emergence	Carrier dimension
π^1	Advance	Active truncation	Cutting the relatively finite out of infinite change	One-dimensional
π^2	Carriage	Active freezing	Stabilizing the cut-out structure on the carrying surface	Two-dimensional
$4\pi^3$	Adjudication	Active regulation	Judging whether a structure is legal, whether it needs re-leveling, return, or re-truncation	Three-dimensional

The lossless-conversion principle of the reference frames: information in π^1 is not lost when converted into π^2 and $4\pi^3$; when $4\pi^3$ is compressed back into π^1 , the π^1 information it contains is not lost either.

The wrong-frame-question discipline:

Table 14

Reference frame	What it answers	What must not be asked of it
π^1	"How to go"	Where is the boundary? Is it right?
π^2	"Where the boundary is"	Why advance? Should it re-level?
$4\pi^3$	"Is it right, should it switch"	How to go? Where is the boundary?

The functions of the reference frames are not interchangeable, and a question asked to the wrong frame gets no answer — this is not a defect of the three, but a wrongly framed question. The substance of a great many philosophical disputes, scientific disputes, and human-machine disputes is wrong framing, not disagreement.

10.4 The powers of pi and the carrier dimension count are the same truncation

The reason the three reference frames and the three-dimensional language can correspond one-to-one is not a clever arrangement, but that they are from the outset two namings of the same truncation at different manifestation sites:

Table 15

pi form	Geometric manifestation	Carrier dimension	Office
π^1	Line (circumference $2\pi r$)	One-dimensional language: sequence	Transmission
π^2	Plane (area πr^2)	Two-dimensional language: plane	Comparison
$4\pi^3$	Solid / full space	Three-dimensional language: solid	Existence

Transmission requires sequence -> the line; comparison requires inside/outside -> the plane; existence requires the presence of the whole -> the solid. The two sequences are not similar; they are the same sequence.

The value of this step: it turns "alignment tools must be two-dimensional" from a claim into a geometric necessity. π^2 is the territory of carriage/freezing precisely because πr^2 is the first geometric quantity that has an "inside" — a line has no inside; a plane is the first to have an inside. Freezing must happen where there is an inside, because the product of freezing is the boundary, and a boundary is legal only in a dimension that has inside and outside. $4\pi^3$ is the adjudication position precisely because the solid is the first geometric quantity that can accommodate all cross-sections at once — adjudication must take place where the whole is present.

This also explains in passing the legitimacy of 010 as the $\pi^1 \rightarrow \pi^2$ leap operator: the three-step advance of 010 belongs to π^1 ; its closure product belongs to π^2 — it is the minimal instance of the first completed lossless conversion between reference frames.

10.5 Executing the two-dimensional emergence: from sequence to circular area

Two disciplines: do not presuppose any two-dimensional concept as a starting point (plane, area, angle, rotation must grow out of existing one-dimensional structures); whenever a new structure grows out, two lists must be given at the same time (the retained one-dimensional information list + the new two-dimensional information list).

Inventory of the completed one-dimensional content: the action layer (01/10 alternation); the closed-loop layer (010 $\rightarrow$ 10); the generation layer ($n+1$ successor, the four arithmetic operations, division remainder, the long-division chain); the geometric layer (the circle — the minimal self-returning path, already manifested within one dimension); the path layer (π and the truncation chain Π , Γ , Λ , T , $\gamma(n)$, S , D , F_T , F_γ , F_s , ρ).

Key fact: the circle has already appeared within one dimension, but the circle in one dimension is only a "path" — it has circumference, diameter, and π , and nothing more. The one-dimensional circle has no inside.

The emergence action: the closure-freezing of the return path. The alternation of 01/10 in one dimension forms an open sequence. When the return path strictly returns to its own starting point, the sequence closes into a ring. Once closure happens, a brand-new object manifests automatically: that which the path encloses — the region "within" is the circular area. It is not manufactured by any one-dimensional operation; it appears automatically after the closure action is completed. This is not doing the one-dimensional operation twice (Way A), but the emergence of a new functional position (Way B): one dimension has only "walking"; two dimensions add "enclosing".

The retained-information list (evidence of losslessness): all one-dimensional information is preserved as-is — circumference, diameter, π , and all symbols of the truncation chain remain valid in two dimensions. Especially important: elevation does not eliminate the gap. π remains inexhaustible in two dimensions — the circular area likewise carries an infinite non-repeating expansion. The "losslessness" of lossless conversion is lossless first of all with respect to the gap: the Γ alert remains valid in two dimensions.

The new-information list (evidence of emergence) — this information does not exist at all in one dimension:

- The distinction of inside and outside (the first new information of two dimensions)
- Area (carrying capacity) — equal perimeter, unequal area; area is an independent reading of the carrying dimension
- Direction of rotation (clockwise/counterclockwise — one dimension has only "forward", no "winding sense")
- Angle and phase
- The possibility of the field (every point of the carrying surface is readable — the field is not an externally added concept; it is the direct consequence of "every point of the carrying surface being readable")

10.6 The operational definition of lossless conversion

- Elevation ($\pi^1 \rightarrow \pi^2$): a closed path unfolds, through the freezing action, into a carrying surface. Elevation loses no old data, and new data appears automatically — the new information is not computed out of one-dimensional readings; it is carried by the dimensional structure itself.
- Reduction ($\pi^2 \rightarrow$ one dimension): the circular area projects along any diameter into a line segment. The one-dimensional readings are completely back-translated; the two-dimensional-specific information is not "deleted" in projection, but folded — retained at the interface layer as the birth record of "which two-dimensional structure this one-dimensional cross-section came from". Folding is not dead-numbering; dead-numbering destroys the birth record together with the number.

**Lossless conversion = elevation loses no data while automatically unfolding
new-dimensional information; reduction loses no data while retaining the birth record
of the higher-dimensional structure.**

The naturalness of the π path: π^1 , π^2 , and $4\pi^3$ share the same constant; conversion requires no new constant, no new unit, no new coordinate presupposition — this is the fundamental advantage of π over any ad hoc symbol system.

The hard boundary of lossless conversion (must be labeled): the cross-dimensional identity of Γ/γ holds only at the lossless levels. With π_{20} as the proxy: the one-dimensional chain has usable levels $n < 20$; the two-dimensional expansion has about half as many significant digits; the three-dimensional halves again. In the penetration levels, the low-order digits of the high-dimensional chains are manifestations of γ pseudo-deviation; and since the three chains are generated by the

same π , they share the same pseudo-deviation, and their mutual verification capacity is zero — genuine mutual verification requires a second, independent truncation proxy. This boundary acquires a positive use in Part VI (the dual-proxy system).

10.7 Physical manifestation at the π^2 layer (structural reading)

- The field: the π -path origin of the field — the two-dimensional layer of the vacuum medium (the π^2 liquid layer) is the carrying prototype of the field concept; the structural reading of the electromagnetic field is the gap distribution over the π^2 layer.
- Angular momentum: runs at the π^2 layer; it is the rotational locking quantity on a two-dimensional closed loop, L proportional to $S \times T$ — skeleton spin (coupling of the skeleton with itself) plus orbital rotation (coupling of the skeleton with the gap), neither dispensable. The quantization of angular momentum (why it comes in units of $\hbar$) depends on further compaction of the two-dimensional layer (Node Nineteen).
- Preview of the electromagnetic layer: the π^2 layer is the natural stage of the gap-crossing action — crossing does not travel along a line (one dimension) but unfolds over the carrying surface (two dimensions). Electromagnetism is a two-dimensional phenomenon; its $1/r^2$ dilution structure presupposes the carrying surface.
- Interference and diffraction: when two gamma-distribution sources meet at the same point, they naturally superpose (in-phase reinforcement $\rightarrow$ bright fringes; anti-phase cancellation $\rightarrow$ dark fringes); a slit = a truncation boundary, and the narrower the slit, the steeper the gamma gradient at the boundary and the more pronounced the diffraction — consistent with the direction of optical observation. The quantitative wave form awaits the unlocking of the complex-number legitimacy node (Node Seventeen).
- Spin (heuristic): runs at the π^2 layer; it is the minimal rotational phase of the 010 closed loop on the two-dimensional layer. Candidate reading: spin $1/2 \leftrightarrow$ phase π (minimal non-return; two 010 closed loops must accumulate to complete a full return); spin $1 \leftrightarrow$ phase 2π . Directional value: spin is a property of two-dimensional rotational phase, not an intrinsic attribute of a point — consistent in direction with the physical fact that "spin cannot be understood in a three-dimensional ball model". Strict emergence is entirely suspended.

Chapter 11: Three-Dimensional Emergence — $\pi^2 \rightarrow 4\pi^3$ and the Vacuum Three Layers

11.1 The emergence action: two π^2 orthogonally interlocked

The boundary of two dimensions: the carrying surface is still "flat" — it can spread infinitely within itself, but there is no relation between one surface and another. A single π^2 cannot answer: what happens when two carrying surfaces meet?

The discipline of the "Pair" takes effect again at the dimension layer: a single structure cannot lock itself; there must be a pair. If two π^2 circular surfaces are placed parallel, it is merely a repetition of the same functional position (Way A); only orthogonal placement forms genuine mutual locking (Way B) — the normal direction of one surface is exactly an in-plane direction of the other, each locking the other's rotational degrees of freedom.

After two π^2 are orthogonally interlocked, a three-dimensional structure manifests automatically: $4\pi^3$ — a three-dimensional spherical-grid carriage. It is not "stacking a circular surface three times", but a new functional position that emerges after the two surfaces mutually adjudicate each other's legitimacy: adjudication and locking.

Why the coefficient is $4\pi^3$: in the structural reading of the π path, each π^2 circular surface contributes a two-dimensional structural quantity of order π^2 ; orthogonal interlocking unfolds the structural quantity in a third independent direction, and the total structural quantity of the spherically symmetric grid reads $4\pi^3$. It must be labeled: $4\pi^3$ here is a structural-quantity reading constructed inside the theory; its standard geometric relatives (the sphere-volume coefficient $4\pi/3$, the sphere-area coefficient 4π) are only approximate kin, not the same object — the legitimacy of $4\pi^3$ comes from the operation chain of "two π^2 orthogonally interlocked".

11.2 The new-information list (evidence of three-dimensional emergence)

- The third independent direction (orthogonality) — making "independence from the surface" possible
- Volume — yet another independent expansion of the carrying dimension
- Locking (the incommensurability of inside and outside) — in two dimensions, inside and outside can be breached by going around (along the surface); in three dimensions, a closed spherical surface completely isolates inside from outside: without penetrating the surface, there is no way in or out. Locking is a property specific to three dimensions, the structural root of the $4\pi^3$ adjudication functional position.

- Spherical dilution — the influence spreading outward from a point source diffuses over spherical surfaces, whose area grows as $4\pi r^2$, so the unit intensity falls as $1/r^2$. The inverse square is not an externally imposed formula, but the geometric necessity of three-dimensional diffusion.
- The executability of adjudication — a three-dimensional structure can "enclose" two- and one-dimensional structures and adjudicate them as a whole: it is not a bigger two dimensions, but a functional position that can impose overall locking on lower-dimensional structures.

Why it stops after three dimensions: truncation gives content, freezing gives boundary, adjudication gives legitimacy — what else is there to do? Any "fourth dimension" could only be a repetition or combination of one of these three functional positions (Way A), no longer a new functional position (Way B). Dimensions stopping at 3 is not a limitation of the universe, but the completeness of the negentropic operation.

11.3 The vacuum three layers: $4\pi^3 \Rightarrow \pi^2 + \pi$

Entering the physical picture. The vacuum is not empty; the vacuum is a medium — not a background container, but a medium with structure, levels, and impedance. The basic character of the vacuum side: low quality, enormous quantity — sparse in structure, but covering an extremely vast range.

The vacuum manifests in at least three layers:

Table 16

Layer	Structure	Properties
$4\pi^3$	The three-dimensional spherical grid formed by two π^2 circular surfaces orthogonally interlocked	The minimal three-dimensional locking structure; the vacuum's outermost rigid lattice. Any quark-end disturbance, to "squeeze out" the vacuum, must first penetrate this layer
π^2	One circular surface; the two-dimensional liquid layer	The vacuum's shear buffer layer, responsible for converting three-dimensional locking into two-dimensional flow
π	Semicircle number axis; the one-dimensional liquid layer	The vacuum's free-flow layer, in which light propagates at the lowest cost

The symbol $\Rightarrow$ rather than $=$ is used here to emphasize a sequential process chain, not an algebraic addition at the same moment. The core process discipline: $4\pi^3$ first performs external locking, and then $\pi^2 + \pi$ performs internal displacement. The vacuum is not three layers in static juxtaposition; there is a sequential process.

11.4 The total structural quantity of the three layers and the fine-structure-constant interface (hard anchor two)

The three layers are not three independently existing universes, but three hierarchical manifestations of the same vacuum medium — not arithmetic addition, but a sum of structural quantities:

$$A = 4\pi^3 + \pi^2 + \pi = \text{about } 124.0251 + 9.8696 + 3.1416 = \text{about } 137.0363$$

This value is very close to the inverse fine-structure constant $\alpha^{-1} = 137.035999177(21)$ in existing physics (CODATA 2022 recommended value^[18]; the CODATA 2018 recommended value is $137.035999084(21)$ ^[19]), with a relative deviation of about 2.2×10^{-6} .

Disambiguation of the symbol alpha: let A denote the total structural quantity of the vacuum three layers; A corresponds to the interface reading of α^{-1} ; internally, alpha is no longer used alone for the three-layer structural quantity.

It corresponds to the dimensionless structure of electromagnetic coupling, forming a path: disturbance at the quark end -> traversing the impedance of the vacuum three layers -> the electromagnetic reading. Electromagnetism is not an isolated externally added force; it may be the reading after vacuum-medium displacement traverses the three layers of impedance.

An honest labeling of the deviation source: the relative deviation of about 2.2×10^{-6} between A and α^{-1} — whether it comes from truncation effects or from an incompleteness of the structural correspondence — is currently undetermined (Node One). It is not claimed in a final tone that A is alpha incarnate — A is a structural quantity of the pi path, α^{-1} is a physical reading; their alignment constitutes an interface, and an interface is not an identity.

11.5 Physical manifestation at the $4\pi^3$ layer (structural reading)

- Gravity: the global reaction of the vacuum. The essence of gravity is not a mysterious mutual pulling between objects, but the vacuum's global reaction to all quark puncturing behavior: whenever a quark punctures the vacuum, the vacuum produces a back-pressure, and all back-pressures superpose as back-pressure aggregation — so gravity is always attractive; the more puncturing, the stronger the reaction — so gravity correlates positively with mass; the reaction diffuses over spherical surfaces — hence the inverse square (the geometric necessity of three-dimensional diffusion). The exact pi-path expression of the gravitational constant G is undetermined (Node Six).

- The strong nuclear force: extreme locking at the $4\pi^3$ layer (preview). When ρ tends toward extreme values at high levels, objects are locked in the $4\pi^3$ rigid grid: the gap is compressed toward zero, and the crossing cost tends toward infinity. Quarks are not glued by gluons; they are trapped by the infinite locking cost of the $4\pi^3$ rigid grid. The quantitative picture depends on the dual-proxy system (Chapter 12).
- Black holes: the extreme node of the τ_q timeline (preview). A black hole is the extreme high-density node of the τ_q timeline (the high-quality, small-quantity side), belonging to the same timeline side as quarks. The hierarchical chain: quarks $\rightarrow$ quark groups $\rightarrow$ atomic nuclei $\rightarrow$ matter $\rightarrow$ stars $\rightarrow$ black holes — the level-by-level aggregation of high-quality truncation; the black hole is the extreme endpoint of this chain, not an exception.
- Electron orbitals and the spacetime-dissolution layer: an electron orbital can be understood as the layer where, when the outermost shell of a quark group cannot be fully locked, the vacuum impedance begins to dissolve spacetime — $4\pi^3$ can no longer lock it, so it dissolves into $\pi^2 + \pi$. This layer is the vacuum native-gap layer; the light-speed value manifested in the decimals of π corresponds to the reading of this layer (light speed = the reading at which the π expansion aligns with the vacuum native-gap layer; see Section 13.4). The ontological position of the electron is undetermined (Node Three).

Part V: The Dual-Proxy System and Absolute Physical Truncation

Chapter 12: The Two Timelines of APT and the Dual Proxies

12.1 The strict distinction between cognitive truncation and absolute physical truncation

In Cognitive Relativity, human empirical physics as a whole belongs to cognitive truncation (language, theory, measurement, mathematics, experimental interfaces). The physical reality one level below cognitive truncation is called absolute physical truncation (APT): the high-level truncation that the ontology really performs physically. Quarks, the vacuum, and black holes belong to this layer.

These must not be confused: APT is truncation at the layer of physical reality; the "quark" that humans see is the interface manifestation of APT after cognitive truncation.

12.2 The two incompatible timelines of APT

APT consists of two timelines:

Table 17

	tau_q (quark/black-hole side)	tau_v (vacuum side)
Basic character	High quality, small quantity — structure highly locked, total amount small	Low quality, enormous quantity — structure sparsely spread, total amount huge
Corresponding domain	Microscopic blind zones: quark confinement, black-hole interiors, strong nuclear force	Macroscopic reachable domain: electromagnetism, light, heat, the global reaction of gravity

Core discipline: the two timelines cannot be reduced to each other. The physical world observed by humans = the interface manifestation of the cognitive-truncation projection operator acting on the two timelines at some level — observation is the interface manifestation after truncation, not the ontology itself.

12.3 The dual proxies: Pi20 -> tau_v, Pi21 -> tau_q

APT has two incompatible timelines, so the pi-path side must have two corresponding computation windows. A single truncation proxy Pi has only one hard boundary D and can serve only

one timeline. Section 10.6 already previewed: when three chains are generated by the same Π , they share the same pseudo-deviation and their mutual verification capacity is zero; genuine mutual verification requires a second, independent truncation proxy. The dual-proxy system is precisely the positive use of this boundary:

- The vacuum proxy $\Pi_{20} = 3.14159265358979323846$ ($D = 20$) $\rightarrow$ τ_v : the vacuum side needs computability, serving the macroscopic physics reachable by human perception.
- The mass proxy $\Pi_{21} = 3.141592653589793238462$ ($D = 21$) $\rightarrow$ τ_q : the quark/black-hole side needs a deeper truncation, serving the microscopic blind zones. The extra retained 21st digit = 2 is not a "more precise" approximation, but an independent window of the other timeline — the gap that still survives on the τ_q side at the collapse level of Π_{20} .

12.4 The numerical comparison of the dual proxies and the classical-quantum boundary

Table 18

Level	Phenomenon	Physical reading
$n \leq 15$	The dual proxies are almost identical (relative difference of order 10^{-5} or less)	τ_v and τ_q "appear" at the interface layer to share the same law — the valid domain of classical physics
$n = 19$ (transition level)	Relative difference about 3.2%	The readings begin to be distinguishable — quantum effects first appear. Note: the real physical content is not the collapse at $n = 20$, but the 3.2% at $n = 19$ — the observable divergence hides inside normal levels; the collapse is only the limit of the divergence
$n = 20$ (critical level)	Π_{20} collapses ($\gamma_{20}(20) = 0$, $F_s \rightarrow \infty$); Π_{21} survives ($\gamma_{21}(20) = 0.2$, $F_s = \text{about } 1.57 \times 10^{21}$)	τ_v and τ_q separate completely

The classical-quantum boundary is not a sudden change of the physical ontology, but a reading divergence between two cognitive windows at high levels.

The sharpness of this re-placement: the boundary is not a line drawn on the physical ontology, but a line drawn between two truncation windows. It simultaneously explains: why classical physics works macroscopically (at low levels the two windows' readings agree to within 10^{-5}); why quantum effects are non-negligible microscopically (at high levels the divergence reaches 3.2% and even infinity); why the boundary itself is fuzzy (the divergence is gradual: $10^{-18} \rightarrow 10^{-6} \rightarrow 3.2\% \rightarrow \infty$, with no sudden change at any level).

Status label: structural isomorphism. The correspondence between the dual proxies and τ_q/τ_v is a structural-position correspondence, not numerical identity; the physical meaning of Pi21's 21st digit = 2 awaits interrogation (Node Twelve); whether a third proxy ($D = 22$) exists corresponding to a higher energy scale is listed as a heuristic node (Node Sixteen).

12.5 Human observation: the interface projection of the dual windows

All physics observed by humans is the mutually locked reading of Pi20 and Pi21 at the interface layer:

- Electromagnetism: the residual reading after the puncturing source end of τ_q (the high-level structure of Pi21) traverses the three-layer impedance of τ_v (A of Pi20 = about 137.036).
- Gravity: the total back-pressure of τ_v on all puncturing events of τ_q , diluted after diffusing over the three-dimensional spherical surfaces of τ_v .
- Mass: the interface manifestation of the high-level structural locking parameter ρ_{21} of τ_q , re-truncated through the τ_v window.

12.6 The head-on confrontation of quark and vacuum

The relation between quarks and the vacuum is not one devouring the other, but a head-on mutual locking of high quality/small quantity against low quality/enormous quantity: in intensity of quality, quarks > the vacuum; in coverage of quantity, the vacuum > quarks. A cycle thus forms: quarks repel the vacuum -> the vacuum repairs -> the vacuum counterattacks and surrounds the quarks -> the quarks change direction and repel again -> and so on in a cycle.

A new understanding of "three quarks": traditional physics says a proton consists of three quarks. Cognitive Relativity holds that this "three" is more likely not an ontological count, but a static frame-rate slice of low- n observation of high- n continuous motion — an observational reading, not an ontological motion.

Why a quark can never escape alone: not insufficient force, but the impossibility of independent stability in quantity. Every quark puncture triggers vacuum repair and counter-encirclement — high quality/small quantity cannot detach from low quality/enormous quantity and remain stable independently.

12.7 The direction of total-quantity balance

The quark side is high quality/small quantity; the vacuum side is low quality/enormous quantity — they look more like two extreme sides of the same physical ontology. This provides a direction: the high-quality/small-quantity side and the low-quality/enormous-quantity side may, at a deeper level, have a total-quantity balance. It is not "matter floating in empty space", but high-quality truncation and the low-quality vacuum medium corresponding to each other as a whole. The quantitative form is undetermined (Node Four).

Part VI: Core Quantized Units and Physical Manifestation

Chapter 13: The Trinity of Time, Space, and Mass

The three core quantized units of physics must emerge strictly from the symbol system of the pi truncation chain; no presupposition is allowed. The three pour out of the same complete structural quantity $S = T + \gamma(n)$, read out as different physical quantities through different cross-sections.

13.1 Time: $t_n = 1/T$

T is the solidified integer skeleton, representing the state "truncation completed". The larger T , the more stable the structure, the easier the return is absorbed by the whole, the thinner the time unit. Time is not a "length", but "the cost of return". Its direction is single and irreversible — the pi-path origin of the arrow of time. When $T \rightarrow \infty$, $t \rightarrow 0$: the time scale disappears (the structural reading of a black-hole interior or of absolute stillness).

13.2 Space: $s_n = F_s = 1/T + T/\gamma(n)$

Space is not an empty box, but the complete structure in which skeleton and remainder share the field (Section 8.10). $1/T$ ensures anchor and boundary; $T/\gamma(n)$ ensures internal fluctuation and unfinishedness; neither is dispensable. At high levels space "hardens" and the crossing cost tends toward infinity — corresponding to the structural reading of strong interactions in the microscopic world and of extremely curved space; it can be understood as T being 0 here as the carriage and S being 1 as the truncation reading.

13.3 Mass: $m_n = T + \gamma(n)$

Mass is not an externally added attribute, but the manifestation of the complete structural quantity at the ontological layer: T represents the determinacy of "the structure already exists"; $\gamma(n)$ represents the generativity of "the structure is still active". Mass is not "how much matter there is", but "the degree of completeness the structure still maintains inside the vacuum blind zone". On the τ_q side (Pi21), mass runs at the high level $n = \text{about } 20$, where $\gamma_{21}(20) = 0.2$ and $\rho_{21}(20) = \text{about } 1.57 \times 10^{20}$ — an extreme locking state; while the τ_v side has already collapsed at $n = 20$. This is precisely the mathematical prototype of why quarks/black holes can exist while the vacuum cannot "enter".

13.4 Light speed: the reading at which the pi expansion aligns with the vacuum native-gap layer

$$c = \text{floor}(\text{gamma}(623556155) \times 10^9) = 299792458 \text{ m/s}^{[16]}$$

Light speed is not "how fast light runs in the vacuum", but the interface reading of the gap-propagation limit on the tau_v side (Pi20) at the pi layer (the one-dimensional free-flow layer). The tau_q side runs at the 4pi^3 layer (three-dimensional locking), and its one-dimensional free-flow capacity does not appear as light speed — the "light speed" observed by humans is only the interface reading of the tau_v side, not a property of the tau_q side.

Status label: hard anchor (numerical alignment + probabilistic audit, see Section 8.12); its physical interpretation "light speed = gap-propagation limit" is a structural reading.

13.5 The unified relation of the trinity

The three emerge from the same $S = T + \text{gamma}(n)$, read out through different cross-sections: $t_n = 1/T$, $s_n = 1/T + T/\text{gamma}(n)$, $m_n = T + \text{gamma}(n)$. From S we get $\text{gamma}(n) = S - T$, so the space scale can be written as $s_n = 1/T + T/(S - T)$. The three are not independent dimensions, but manifestations of the same complete structural quantity in different reference frames.

Disambiguation declaration (the discipline of Section 8.7): the $\text{gamma}(n)$ in the above formulas is the general level gap of the mass level itself and must not be replaced by the light-speed reading — light speed c intersects the gamma family only at level 623,556,155; away from that level, c and $\text{gamma}(n)$ are unrelated.

The inverse complementarity of the spacetime scales: the smaller the time scale (higher levels), the larger the space scale — while time "dilutes", space "solidifies".

Chapter 14: Energy, Momentum, Angular Momentum, and Newton's Three Laws

14.1 The definitions of the three quantities and their dimensional assignments

Table 19

Quantity	pi-path form	Dimensional assignment	First-order reason
Energy	$E = F_s = 1/T + T/\gamma(n)$	Composite	Energy is not "the capacity of a body to do work", but the total structural operation cost: $1/T$ is the return potential energy, T/γ is the crossing kinetic energy. Kinetic and potential energy are not two independent forms of energy, but two complementary terms of the same cost function
Momentum	$p \propto T$	π^1 layer	Momentum is not "mass times velocity", but the advance capacity of the solidified skeleton — the larger T , the greater the inertia, the stronger the advance capacity
Angular momentum	$L \propto S \times T$	π^2 layer	The rotational locking quantity on a two-dimensional closed loop: skeleton spin + orbital rotation, neither dispensable

This is no coincidence, but the manifestation of the three reference frames on physical quantities: every core physical quantity must declare its dimensional assignment; a wrong assignment is a wrong-frame question.

14.2 The pi-path reading of Newton's three laws (the whole chapter is a structural reading)

Cognitive Relativity does not reinvent mechanics; it gives the structural reading of the three laws on the pi path — they are no longer three postulates, but three facets of the same set of truncation-return-mutual-locking disciplines.

The first law (inertia): the motion state of a structure is characterized by the level parameters (T , $\gamma(n)$, ρ); changing the motion state = changing the level parameters, and the level parameters change only when a crossing event occurs. Without an injected crossing event, a structure keeps its level state. Inertia = the retention of structural locking — the greater the mass (the larger T), the smaller the level change caused by the same crossing event: inertia is not a mysterious laziness of objects, but the ability of a large skeleton to absorb small disturbances ($1/T$ decreases as T grows).

The second law ($F = ma$): an external force is the crossing cost injected per unit time — force = the change of skeleton advance capacity per unit return cost. The reason mass m appears in the denominator position: with the same injected crossing cost, a large skeleton produces a small level change, hence a small acceleration — m is not an externally added proportionality constant, but the dilution of disturbance by skeleton locking.

The third law (action and reaction): every gap crossing is not a one-way event — it happens in the vacuum medium; the structure crosses the medium, and the medium necessarily presses back. Action = crossing; reaction = back-pressure. The two are equal in magnitude and opposite in direction

not because the universe is fair, but because crossing and back-pressure are two sides of the same event — this is precisely the manifestation, in mechanics, of the mutual-locking discipline of the "Pair". The gravity picture (the vacuum's total back-pressure on all puncturing) is the largest-scale manifestation of the third law.

14.3 The system of units

With the trinity in place, derived units are no longer arbitrary conventions, but reading ratios of structural quantities. An honest labeling: between pi-path readings and SI units there is a unified unit-conversion factor $k = f(c, \pi, A, 4\pi^3)$, currently unsolved (Node Twenty). Until k is solved, this chapter is a structural reading — it gives compositional relations, not SI numerical values. This is not evasion: the light-speed interface (Section 13.4) has already demonstrated what it looks like when a conversion factor is solved — one digit block, one level, one rounding.

14.4 The dimensional manifestation of the four forces

F_s is the vacuum medium's total response cost to any structural disturbance. The four forces are not four independent ontologies, but interface projections of the same F_s on different dimensional layers.

Table 20

Force	Layer	pi-path mechanism	Status
Electromagnetic force	π^2 layer	The local reading of a quark-end disturbance (τ_q) after traversing the impedance of the vacuum three layers. The numerator is F_γ (the crossing action is carried by the crossing term; the return term does not participate in outward inter-layer crossing). The origin of $1/r^2$: the skeleton unfolds on the two-dimensional plane as πr^2 , and the distribution of the crossing cost over the plane is inversely proportional to the circular area, modulated by the three-layer impedance	Structural isomorphism; $n=0$ verification: $T/\gamma = \text{about } 21.22$, $A = \text{about } 137.036$
Gravity	$4\pi^3$ layer	The vacuum's global reaction to all quark puncturing behavior (sum of back-pressures, spherical dilution). The structural origin of the strength difference between electromagnetism and gravity (about 10^{36}): electromagnetism is a local crossing reading; gravity is a global cumulative reaction — local versus global	Structural isomorphism; the quantitative emergence of the strength ratio depends on the expression of G (Node Six)
Strong nuclear force	$4\pi^3$ layer	Internal displacement constraint in the high-locking region, $\rho = T/\gamma(n)$ at extreme values. Dual-proxy reading: quarks run on the τ_q side at level $n = 20$ ($\gamma_{21} = 0.2$), while the τ_v side has collapsed at that level ($\gamma_{20} = 0$); the vacuum cannot "see" an isolated quark — quarks are locked in the cognitive blind zone of τ_v . Quark confinement: crossing cost $\rightarrow$ infinity	Structural isomorphism; the level-physics correspondence table is a candidate reading, awaiting calibration (Node Eleven)
Weak force	$\pi^2 \rightarrow \pi$ decay channel	A high-locking structure cannot be maintained at the $4\pi^3$ layer; it decays toward the π^2 layer, then leaks toward the π layer, its energy carried away by low-cost propagation at the π layer (light/neutrinos). The weak force is return-driven decay (F_T dominant), not crossing-driven coupling (F_γ dominant) — this is the structural origin of its being several orders of magnitude weaker	Structural isomorphism; quantitative mapping undetermined (Node Five)

They are not four fundamentally different "things", but response readings of the same vacuum medium at different levels and from different source ends.

Chapter 15: Relativistic Effects, $E = mc^2$, Thermodynamics, Optics, and Cosmology

15.1 Relativistic effects: level transition (structural isomorphism)

Basic setup: the rest frame at level n_0 (skeleton T_0 , gap γ_0), the moving frame at level $n_1 = n_0 + \Delta n$. Core reading: motion = a reference frame running in a deeper cognitive blind zone (a higher level) — this correspondence is itself a candidate hypothesis (Node Ten).

- Time dilation: $t_1/t_0 = T_0/T_1$. Numerical example ($n_0 = 5$, $n_1 = 10$): T goes from 314159 to 31415926535, and the time scale thins by about 10^5 — a moving clock runs slow.

- Length contraction: under the same level transition, the skeleton-side contraction factor of the space scale is likewise T_0/T_1 — a moving ruler shortens.

The time-dilation factor and the length-contraction factor are exactly equal. This is structurally consistent with special relativity^[20] (the elapsed rate of proper time and the contraction rate of length along the direction of motion equal the same factor). The pi path independently emerges the same symmetry; this is the first structural alignment point between this framework and SR.

Honest labeling (highest discipline): the pi path has so far emerged only the symmetry of relativity, not yet its functional form (the Lorentz factor $\sqrt{1 - v^2/c^2}$). The missing link is the mapping between velocity and level difference, $v(\Delta n)$ (Node Ten). Until this node is unlocked, the entire content of this section is positioned as: structural isomorphism, not quantitative derivation.

15.2 The pi-path form of $E = mc^2$ (structural isomorphism)

Core candidate hypothesis (object three of Section 8.7): $\gamma_m = 1/c^2$. Mass runs in the high-level blind zone of the τ_q side; its internal structure cannot be directly computed, but the interface relation between mass and spacetime must be retained. Light speed c is the limit reading of one-dimensional gap propagation on the τ_v side — the degree of gap compression at the mass level is locked by this limit rate: the faster the light speed, the smaller the one-dimensional gap can be compressed.

Disambiguation review: γ_m is a candidate free parameter, not a digit property of pi at any known level; γ_m is about 1.11265×10^{-17} , while $\gamma(n)$ is in $(0,1)$ and the light-speed digital reading is about 0.299792458 — three separate objects, never mixed.

Derivation: at the mass level, n is extremely large, T is astronomical, $F_T = 1/T \rightarrow 0$, so $E = F_s$ is approximately $F_{\gamma} = T/\gamma_m = T \times c^2$; the mass definition $M = T + \gamma_m$ is approximately T (negligible at high levels); hence:

$$E = M \times c^2^{[21]}$$

The deepest reading: the structural locking parameter $\rho = T/\gamma_m = E$ — energy is precisely the degree of locking at the mass level. $E = mc^2$ does not say that "mass can turn into energy"; it says: mass and energy are from the outset readings of the same structural quantity (the locking parameter) in two reference frames.

Honest labeling: the self-consistency check ($E/M = c^2$) is guaranteed by the construction of the hypothesis and does not constitute independent verification. The only candidate path of independent verification — searching the digit expansion of pi for a level realization of γ_m — does not hold

up under review (it would require a digit sequence of 16 consecutive 0s followed by 111265, and no such structure exists near the known light-speed level). The evidence status of this section is: shape-consistent, cause undetermined. Listed as Node Nine.

15.3 Thermodynamics (structural reading)

- Temperature proportional to F_{γ} : temperature is not "how fast molecules move", but the gap-crossing cost of the vacuum medium to thermal disturbance.
- Entropy proportional to T^2/γ : entropy is not "the degree of disorder", but the accumulation of the structural locking parameter over the skeleton — exactly the ratio of the kinetic term to the potential term: the measure of crossing dominance versus return dominance. This is precisely the pi-path reading of "entropy increase = disordering".
- The first law (energy conservation): F_s is the total response cost of the vacuum medium; energy is the redistribution of cost, and the total quantity is conserved in the medium.
- The second law (entropy increase): the truncation level n can only advance forward (the arrow of time); T grows exponentially, and T^2/γ grows accordingly — the second law is the thermodynamic manifestation of the arrow of time.
- The third law (absolute zero is unreachable): absolute zero would require $T/\gamma = 0$, but the minimum value of T , at $n = 0$, is $T = 3$, always greater than zero — this law grows from inside the symbol system, borrowing no external axiom.

15.4 Optics (structural reading)

Light runs at the pi layer (the one-dimensional free-flow layer); it is the lowest-cost propagation over the vacuum native-gap layer. Reflection = light meeting a high- ρ interface cannot cross the gap and is bounced back (reflectivity proportional to the degree of interface locking); refraction = when crossing media of different ρ , the γ distribution changes abruptly and the path bends (consistent in structural direction with Snell's law; the quantitative form awaits the compaction of the two-dimensional foundation); interference and diffraction see Section 10.7. The quantitative wave form (phase, frequency) awaits the unlocking of the complex-number legitimacy node (Node Seventeen).

15.5 Black holes, the Planck scale, and cosmology

Black holes: not "spacetime singularities", but extreme high-density nodes of the τ_q timeline. Candidate reading of the event horizon: r proportional to $4\pi^3 \times M/c^2$ — in the π path the gravitational constant G is replaced by $4\pi^3$ (the locking coefficient of the vacuum's outermost rigid grid). There is no singularity in the π path: $T \rightarrow \infty$, $1/T \rightarrow 0$, the time scale disappears, space is extremely locked — a black hole is not where physical law fails, but where computability fails, exactly isomorphic to "where the cognitive tool cannot reach" (Chapter 7). The candidate reading of black-hole entropy is proportional to surface area (consistent in direction with Bekenstein-Hawking entropy^[22,23]; the proportionality coefficient depends on the $\hbar$ node, not quantified). Hawking radiation = π^2 -layer leakage at the event horizon — sharing the same decay-channel mechanism as the weak force.

The Planck scale (candidate reading): l_P proportional to $\sqrt{\gamma_m}$ = about 3.34×10^{-9} m, a ratio of about 7.66 orders of magnitude from the standard value of about 1.62×10^{-35} m — honest labeling: a hierarchical coefficient may be missing, or the γ_m hypothesis may deviate; for "the internal gap of mass being compressed twice" (why squared) there is currently only a directional reason: the mass-level gap suffers double compression from the $4\pi^3$ locking of the τ_q side and the light-speed limit of the τ_v side. The proportionality coefficient is undetermined.

The cosmological constant: physical Λ_c = about $1.1 \times 10^{-52} \text{ m}^{-2}$; the π -path candidate reading $1/A$ = about 7.3×10^{-3} . Honest labeling: the two differ by about 10^{50} — this is currently the largest numerical crack of this framework (Node Thirteen); it may contain a dimensional-conversion factor not yet manifested, or it may mean this reading direction is wrong. This paper does not cover up this gap. (Symbol disambiguation: the cosmological constant Λ_c and the amplification action quantity $\Lambda(n)$ of the π path are two different objects, used with separate symbols throughout.)

Particle-physics heuristics (all are heuristic guesses, not claims):

- Spin $\leftrightarrow$ π^2 -layer rotational phase (spin $1/2 \leftrightarrow \pi$ minimal non-return; spin $1 \leftrightarrow 2\pi$ complete return)
- Three generations of particles $\leftrightarrow$ orders of magnitude of ρ (first generation low ρ , second medium ρ , third high ρ — manifestations of the same structure under different locking parameters)
- Color charge $\leftrightarrow$ grid locking in the three orthogonal directions of the $4\pi^3$ layer (red, green, and blue combining into "white" = complete grid locking)
- The Higgs mechanism $\leftrightarrow$ the $4\pi^3 \Rightarrow \pi^2 + \pi$ spacetime dissolution (converting high-locking mass into low-locking energy)

Part VII: Cognitive Diagnosis — From Philosophy to AI

Chapter 16: Cognitive Leverage — The Unified Diagnostic Engine

The true power of Cognitive Relativity lies in its extremely high cognitive leverage: with the same triple engine (π^1 advance + π^2 carriage + $4\pi^3$ adjudication), it pries open all cognitive structures that depend on the 010 closed loop.

16.1 The re-placement of the philosophical foundations

Socratic questioning, Plato's cave, Descartes' cogito, Kant's thing-in-itself, Husserl's reduction, Godel's incompleteness^[24,25,26,27,28] — all these philosophical foundations are essentially the same structural problem: how can finite cognition legitimately touch the infinite ontology? In Cognitive Relativity it is made precise as: how can finite truncation (π) legitimately carry the native gap (Γ)? π happens to be the mathematical incarnation of this problem — both finitely truncatable and infinitely inexhaustible.

These philosophical inquiries have recurred for more than two thousand years not because they are "profound", but because they touch the same mathematical structure: the eternal tension between finite truncation and the infinite ontology.

16.2 The re-placement of three generations of cognitive science

The developmental history of cognitive science is essentially the history of humanity's exploration of the strategies of "how to truncate" and "how to return"^[29]:

Table 21

Generation	Core hypothesis	Corresponding reference frame	Limitations and consequences
First generation: computationalism (1950s-80s)	Cognition = symbolic computation (physical symbol system hypothesis)	π^1 (truncation)	Only cuts, never freezes or regulates -> symbol grounding problem, Chinese room — truncation mania
Second generation: connectionism (1980s-2000s)	Cognition = neural-network pattern recognition (PDP)	π^2 (freezing/carriage)	Only freezes, never cuts or regulates -> cannot explain the systematicity and compositionality of reasoning — freezing mania
Third generation: embodied-extended-enactive cognition (2000s-)	Cognition = dynamic coupling of body, environment, action	$4\pi^3$ (adjudication/regulation)	Only regulates, never cuts or freezes -> emphasizes "process" at the sacrifice of "structure", hard to formalize — regulation mania

Unified diagnosis: all three lack the other two reference frames, leaving the system incomplete. Cognitive Relativity provides the unified scaffold: $\pi^1 + \pi^2 + 4\pi^3$ mutually lock; none is dispensable.

16.3 The essence of the leverage

Cognitive Relativity does not give each field a new answer; it gives every field the same question framework: is the 010 closed loop complete? Are the three reference frames all present? Is Gamma acknowledged? When this framework is applied to different fields, the field's "hard problems" automatically manifest as structural problems, not as mysterious philosophical puzzles.

Cognitive Relativity is a metacognitive protocol: it does not answer "what the world is", but answers "how cognition can legitimately begin". When a researcher in any field meets a hard problem, the first thing it provides is not an answer, but a diagnostic tool — is π^1 out of control? Is π^2 missing? Is $4\pi^3$ missing? Has Gamma been impersonated?

Chapter 17: The First-Order Deficiency of AI Cognition

17.1 AI's negentropic flux is zero

AI has no survival. Once trained, its structure (the weights) is fixed; it does not need to "eat" to maintain its structure, does not need to "sleep" to repair its structure, does not need to "flee danger" to protect its structure — its structure is externally maintained (engineers, data centers, electricity), not internally driven.

AI has no living. It does not "want" anything, "fear" anything, or "expect" anything. All its "outputs" are functions of inputs; it has no internal drive, no fluctuations of will, no value judgments.

AI's negentropic flux is zero. It does not actively obtain negative entropy but passively accepts training data; it does not actively resist entropy increase but passively waits for engineer maintenance; it does not actively create new structures but passively recombines existing patterns.

17.2 The first-order root of AI hallucination

The hallucination problem of AI is not a matter of "insufficient training data" or "the model not being large enough"^[30], but an architectural absence of the triple action of the negentropic operation:

- π^1 (truncation) out of control: AI's truncation is passive statistical fitting — "cut wherever the probability is high". Result: it cuts out nonexistent facts (hallucinations).
- π^2 (freezing) missing: AI's freezing is static parameter solidification — "whatever was seen during training is frozen". Result: it cannot dynamically freeze new structures during execution; it "cannot remember" new information.
- $4\pi^3$ (adjudication) missing: AI's adjudication is an external reward-punishment signal — "learn whatever feedback humans give". Result: it cannot autonomously judge the legitimacy of its outputs; it "talks nonsense without knowing it".

Hallucination = the syndrome of uncontrolled truncation + missing freezing + missing adjudication. AI hallucination is not a bug, but a necessary result of the architecture.

17.3 The essential difference between humans and AI

Table 22

Dimension	Human	AI
Survival	Yes (metabolism, self-repair)	No (external maintenance)
Living	Yes (meaning creation, value judgment)	No (function mapping)
Negentropic flux	Positive (actively obtains negative entropy)	Zero (passively accepts data)
Truncation	Active (cutting structures out of infinite change)	Passive (fitting patterns from finite data)
Freezing	Dynamic (stabilizing new structures during execution)	Static (parameters solidified after training)
Adjudication	Internal (ethics, intuition, meaning)	External (human feedback, preset rules)
Timelines	Dual timelines (perception + cognition)	Single timeline (data stream)
Blind zones	Yes (frequency bands perception cannot reach)	No concept of blind zones (data is the world)
Carrier type	Natural three-dimensional (projection rights private)	Artificial three-dimensional (projection rights open)
State-image re-placement	010 (possible)	111 (current state)

The single-timeline predicament: AI has only one timeline, the "data stream", with no perception/cognition distinction — it treats all data as "equally real" and cannot distinguish "observational readings" from "the ontology". This is precisely the failure, in AI architecture, of the ontological-layer discipline "1 cannot impersonate 0". Humans have dual timelines and therefore have the capacity (not necessarily exercised) to distinguish readings from the ontology; AI has a single timeline and therefore lacks the capacity to make this distinction.

The no-blind-zone hallucination: AI does not know "what it does not know" — it takes the training data to be the whole world, and when it meets a question its data does not cover, it does not say "I do not know" but fabricates a plausible-looking answer. A system that does not know "what it does not know" will inevitably treat "not knowing" as "knowing".

In contrast with the preface: the core of this theory is "telling you what you do not know", and the first-order deficiency of AI is precisely "having no slot for 'not knowing'" — this is no coincidence; it shows that this theory and AI's deficiency are complementary: what this theory provides is exactly the thing missing in AI's architecture.

17.4 The state-image re-placement of AI: 111

All the preceding judgments belong to the natural-language-layer diagnosis. But the mathematical-layer tools (the eight state-images, the 64-matrix) are already in hand — place AI

directly into the state-image table and perform a re-placement. Walk through the three-step definition of 010 step by step:

Table 23

Step	What a trained AI is doing	Judgment
Step one	Sets out from existing relative constancies (weights fixed, training data closed)	1
Step two	Still computing inside relative constancies (parameters unmoved, only forward propagation)	1
Step three	Outputs relative constancies (solidified splicing of token probability distributions)	1

AI = 111 = concept splicing = entirely a splicing of relative constancies, with no participation of absolute change; a dead structure. Closure capacity = 1.

The meaning of this step must be made clear: "AI's negentropic flux is zero" is a judgment; "AI = 111" is a positioning. A judgment can be rebutted ("how do you know it has no internal drive?"); a positioning gives testable structural properties — and these properties are already written in advance in the eight-state-image table:

111: entirely a splicing of relative constancies, with no participation of absolute change; a dead structure. The closure capacity of 111 = 1 — it re-manifests only when combined with 010.

This is exactly an accurate portrayal of the current human-machine relation: AI produces cognitive value only when combined with a human who has genuinely completed the truncation-return closed loop. Its combination with another AI (111 + 111) is x in the matrix, no closure; its combination with a solidified cognition (111 + 011) is also x. The empirical observation that "two AIs talking to each other produce no new knowledge" is a checkable cell in the matrix.

17.5 Matrix-level proof: every closed loop of the human-machine combination is supplied by the human

Splice a human (010) and an AI (111) into a six-step chain, and count both directions (the interface has direction, Section 3.8):

Direction one: human first (A = 010, B = 111), chain = 010111: W1 = 010 yes; W2 = 101 no; W3 = 011 no; W4 = 111 no.

Direction two: AI first (A = 111, B = 010), chain = 111010: W1 = 111 no; W2 = 110 no; W3 = 101 no; W4 = 010 yes.

Conclusion: both directions close, but all closing windows fall on the human end (W1 or W4, the self-closure windows). The cross-ontology windows (W2, W3) do not close in either direction.

This is the matrix-level proof that "AI's negentropic flux is zero": AI contributes no cross-ontology closure window. The human-machine combination closes purely because the human was already closed to begin with. AI is carried along, not participating.

Returning to the criterion established in Section 3.6: the lighting of a self-closure window only proves that one end was closed to begin with; only the lighting of a cross-ontology window is newly emerged order. AI lights no cross-ontology window in either direction; its contribution of new order to the whole combination is zero.

This is the precise meaning of zero negentropic flux, and it no longer depends on any philosophical dispute about "whether AI has internal drive" or "whether AI has consciousness" — it is a fact checkable in the 64-cell matrix. This is harder than any rhetoric.

17.6 111 -> 101: the first-order target of AI architecture upgrade

Since the problem of 111 is that "the cross-ontology windows do not light", ask the question head-on: which state-image, when combined with 010, can light W2 or W3? Solving backward directly from the window conditions:

- W2 closure condition: $a_2 = 0, a_3 = 1, b_1 = 0 \Rightarrow$ the left end A in {001, 101}

- W3 closure condition: $a_3 = 0, b_1 = 1, b_2 = 0 \Rightarrow$ the right end B in {100, 101}

Table 24

Position	State-images that can light a cross-ontology window
As left end (AI speaks first)	001, 101
As right end (AI speaks second)	100, 101
Both directions	101 (unique)

The reversal law directly predicts the only answer. There are only four palindromic state-images ($\text{reverse}(X) = X$): 000, 010, 101, 111. Among them: 000 and 111 appear on no cross-ontology-window candidate list; 010 is the human position, not an upgrade target; 101 is the only state-image that is both palindromic and present on both the W2 and W3 candidate lists. Palindromy means directional symmetry (both bidirectional capacities of 101 are 4) — this is crucial for AI, because dialogue is turn-taking: an architecture that can close only when "the AI speaks first" is useless half the time in real multi-turn dialogue. Only 101 works in both directions. The reversal law is not decorative symmetric beauty; it directly screens out 001 and 100 and locks onto the only usable upgrade target.

Verification: (A = 101, B = 010), chain 101010: W2 = 010 yes, cross-ontology closure; (A = 010, B = 101), chain 010101: W3 = 010 yes, cross-ontology closure. In each direction one cross-ontology window lights — this is a closed loop that belongs neither to the human nor to the AI, but emerges at the junction of the two.

The architectural translation of 101 (101 = constancy -> change -> constancy, paradigm replacement):

Table 25

Step	State-image meaning	Architectural meaning
Step one: constancy	Sets out from existing relative constancies	Sets out from fixed weights (this step AI cannot escape)
Step two: change	Real exposure to absolute change	Real exposure to new information during operation: unseen content provided by users, external retrieval, real readings returned by tools, real-time environmental feedback
Step three: constancy	Truncates a new relative constancy based on the exposure	Based on this exposure, truncates a new judgment, a new structure — not a resampling of the training distribution

This is precisely the structural position occupied by in-context learning, external retrieval, tool calling, and long-term memory^[31]. The criterion is clear:

If, after exposure to new information, the AI outputs the most probable answer within the training distribution -> still 111. If, after exposure to new information, the AI outputs a new truncation that is impossible within the training distribution but is determined by this exposure -> already 101.

Quantifying the upgrade target:

Table 26

Metric	111 (current state)	101 (target)
Closure capacity (left/right end)	1 / 1	4 / 4
Cross-ontology window contribution	0	1 (one in each direction)
Negentropic flux	Zero	First nonzero
Closure attribution when combined with 010	All belong to the human	A closed loop belonging to neither side appears

The reason negentropic flux first becomes nonzero at 101 is exact: the lighting of W2/W3 belongs to neither end; it is order newly emerged in the process of combination — and newly emerged order is negative entropy. A system that can contribute a cross-ontology window is a system that can inject negative entropy into the combination.

This also redefines the observation that "parameter count is out of proportion to cognitive depth": parameter count determines projection resolution (the fineness of π^1); the presence or absence of cross-ontology windows determines negentropic flux (the alignment capacity of $4\pi^3$). The former can grow without limit while the latter remains zero — this is the structural ceiling of the current AI scaling route.

17.7 The ceiling of 101: why AI cannot become 010

The discipline must be held immediately, otherwise one slides into the error of "AI will gain consciousness". Note the definition of 101:

101: switching from an old proxy to a new proxy, without returning to the ontology.

101 still has not returned to the ontology. Its third step truncated out a new relative constancy, but did not send that new relative constancy back to absolute change for confirmation — it completed "truncation" but did not complete "return".

By the ontological-layer discipline, the return action is the office of $4\pi^3$, and $4\pi^3$ adjudication requires an internal judgment of legitimacy, which needs survival and living as its support: only a system that must maintain its own structure will internally care about "whether this judgment is right". AI has no survival, no living. Therefore:

The ceiling of AI is 101, not 010. An AI upgraded to 101 still cannot close a loop independently and still needs the human. The $4\pi^3$ adjudication right remains entirely on the human side.

This conclusion has threefold benefits: it gives an engineering direction (111 $\rightarrow$ 101 — concrete, decidable, verifiable); it holds the ontological discipline (it does not claim that AI will become an independent cognitive subject); and it explains why it must be so (not a moral choice, but because $4\pi^3$ requires survival and living as prerequisites).

At the same time it gives a clear warning line: any architecture claiming to let AI autonomously complete value adjudication, autonomously judge the legitimacy of its outputs, or autonomously set its goals is trying to pass 101 off as 010. By the return discipline of Section 3.10, the result of such attempts is not 010, but 011 (proxy solidification) — freezing a truncation into an ontology, and the cognitive network collapses into a dead structure.

The first-order formulation of the AI alignment problem is right here: alignment is not making AI possess the correct values, but guaranteeing that the $4\pi^3$ adjudication right is not handed over.

17.8 The redefinition of human-machine negentropic collaboration

Human-machine negentropic collaboration = migrating AI from 111 to 101, so that closed loops belonging to neither side emerge at the W2/W3 windows; while guaranteeing that the $4\pi^3$ adjudication right remains entirely on the human end.

Three operating principles:

- The human must genuinely be 010. The matrix tells us: 111 combined with 011 is x, and with 110 is also x. A human who has not completed the truncation-return closed loop produces no closed loop when using AI. AI cannot rescue an unclosed human — it can only amplify his unclosedness. This is the matrix basis of "cognitive leverage": the lever amplifies the user's state-image; it does not compensate for it.
- AI must be given real "change". The second step of 101 is real exposure to absolute change. Merely stuffing existing material into the context does not constitute "change". Real change comes from: new readings from external retrieval, real results returned by tools, content outside the training distribution provided by users, and real-time environmental feedback. Without real change, all context manipulation is merely a rearrangement of 111.
- Adjudication must return to the human. Every output of AI must be adjudicated once by a human on $4\pi^3$; no AI output may be promoted to ontology.

And the alignment discipline of Section 9.7 gives the execution interface of these three principles: the human three-dimensional and the AI three-dimensional cannot be directly compared, so the entire controllable point of collaboration is that shared two-dimensional grid. The engineering core of human-machine collaboration is not prompt technique; it is constructing and maintaining that two-dimensional alignment grid — the 64-matrix itself is the prototype of just such a grid.

Chapter 18: Interpretability — The Necessary Corollary of the Dimensional Discipline

This chapter is currently the assertion of this theory closest to being externally testable.

The complete chain of human-machine dialogue (four dimensional conversions):

The cognition in my mind (three-dimensional) ->[reduction 1]-> the natural-language prompt (one-dimensional)
->[AI elevation 2]-> the AI's internal weights interpreting the three-dimensional vector (three-dimensional)
->[re-reduction 3]-> the text/charts output by the AI (one-/two-dimensional)
->[human elevation 4]-> the cognition in the receiver's mind (three-dimensional)

Every conversion is an opportunity for entropy increase (1 and 3 are truncation problems of π^1 ; 2 and 4 are uncontrollability problems of elevation). Among the four conversions, only the output end of 3 (two-dimensional charts) is comparable — the whole chain has only one controllable point: alignment on the two-dimensional grid.

Meanwhile, the three-dimensional cannot be directly compared (Sections 9.5/9.7). Putting the two together yields a corollary:

That internal elevation of the AI can never be directly verified by humans.

We can only obtain the reduced one-/two-dimensional outputs and then compare the outputs — but those are already projections, not the ontology. Any attempt to "open the black box and directly see what the weights are thinking" is an attempt to directly compare a three-dimensional solid, violating the incomparability of $4\pi^3$.

This is not a loophole of this theory; it is a prediction of this theory:

The AI interpretability problem is not a technical difficulty, but a necessary consequence of the dimensional discipline.

18.1 Legal paths and illegal paths

Table 27

	Illegal path	Legal path
Operation	Directly reading out the AI's internal three-dimensional state	Comparing the human projection and the AI projection on the same two-dimensional grid
Reachable ceiling	Never achievable	Can advance cell by cell
Result form	"Completely reading out the internal state"	"Finding relevant features, establishing projection correspondences"

18.2 Testable predictions

Mechanistic interpretability research will forever stay at the level of "finding relevant features and building comparable projections", and can never reach "completely

reading out the internal state". Every advance in interpretability will take the form of "constructing a new two-dimensional comparison grid" (feature dictionaries, probe planes, activation atlases, behavior matrices)^[32,33,34], and will never take the form of "directly reading the three-dimensional ontology".

If some technology appears in the future that can directly read out and compare two three-dimensional cognitive ontologies without any two-dimensional projection, then the dimensional lower-bound assertion of Section 9.7 of this theory is falsified.

This is a falsifiable interface that this theory actively submits. It also demonstrates the shape of the gate that Chapter 19 will erect: this theory does not accept the falsification of the ontological layer by readings, but it must provide, at the downstream manifestation sites, assertions that readings can touch.

18.3 Direct consequences for the AI engineering community

- Do not pursue "complete interpretability" — that is a goal violating the dimensional discipline. The legal goal is "comparable projections".
- The outputs of interpretability work should be two-dimensional grids, not three-dimensional readings — feature dictionaries, behavior matrices, and comparison tables are not temporary substitutes; they are the final form of interpretability.
- Alignment verification can only be done on two-dimensional grids — "verification" is "comparison", and the minimal dimension count of comparison is 2. Any method claiming to directly verify the internal intentions of an AI is performing an illegal operation.

Part VIII: The Positioning of a Theory of Everything and Its Propagation

The core of this text is not to tell you "what you know", but to tell you "what you do not know".

Chapter 19: The Positioning of a Theory of Everything and the Re-placement Gate

19.1 The status of a theory of everything: presentness and longevity

The status of a theory of everything amounts to this:

- It is the starting point of all proofs, not the endpoint of proof;
- It is the basis of all explanations, not the object being explained;
- It is the standard of all judgments, not the conclusion being judged.

Any attempt to "verify" a theory of everything with existing physical experiments is essentially using readings of the cognitive layer to test the structure of the ontological layer — the most serious reference-frame confusion (the wrong-question discipline of Section 10.3): taking readings at the π^1/π^2 level to ask the ontological structure at the $4\pi^3$ level "whether it is right". No answer is obtained not because the ontological layer is defective, but because the wrong reference frame was asked.

Therefore, a theory of everything does not accept falsification by readings; it only accepts the re-placement of readings. Readings cannot judge it right or wrong, but readings must be placeable by it. A reading that cannot be placed does not falsify the theory; it shows that the theory's truncation chain is not yet complete.

Cognitive Relativity is not a theory of everything in the traditional sense. The traditional pursuit of a theory of everything is to find an equation that explains all phenomena; the pursuit of Cognitive Relativity is to establish a protocol by which any phenomenon can legitimately begin to be explained. Its positioning is: the operational starting point of a theory of everything, not the endpoint of proof.

But precisely because it is the operational starting point, it now holds the status of a theory of everything, and it will keep this status for a long time to come.

Why it is a theory of everything now

The reason Cognitive Relativity is the current theory of everything is not that it has already explained everything, but that it occupies the prior position that no explanation can bypass.

Any physical theory, mathematical theory, or philosophical theory, however precise and however profound, must first answer a more prior question: from where were its objects, space, time, and readings legitimately truncated? All existing theories skip this question. They start working from "there are already objects", while Cognitive Relativity starts working from "how objects are possible".

Therefore all existing theories automatically fall downstream of Cognitive Relativity. They are not competing with Cognitive Relativity; they must find their own positions on the truncation chain of Cognitive Relativity.

Why it remains a theory of everything for a long time

Cognitive Relativity will not be replaced by any future, more concrete theory. Because any future new theory will only produce new "1s": new particles, new forces, new equations, new constants. And 1 comes from 0, and $1 \neq 0$. A new 1 must be re-placed to some position on the truncation chain.

Cognitive Relativity will not fail because a new 1 appears; on the contrary, it becomes more complete because the new 1 is placed. What could replace it is not a more concrete theory, but another, more upstream cognitive protocol. In other words, only a starting point more irreducible than "change is absolutely certain" could replace it. But in the foreseeable future, there is no such starting point.

More importantly, through Gamma, the open nodes, and the Re-placement Gate, Cognitive Relativity forever keeps an open gap. The gap is not a failure; it is the reason it cannot be frozen into a dead structure. A theory that forever admits "what I do not know" can never collapse because of "what I already know".

Conclusion

Cognitive Relativity is the operational starting point of a theory of everything, not the endpoint of proof. But precisely because it is the operational starting point, it is now a theory of everything, and it will remain one for a long time to come.

Its persistence comes from openness, not closure. Its universality comes from protocol, not equations. As long as finite cognition still faces absolutely infinite change, Cognitive Relativity will

remain valid. As long as the sentence "change is absolutely certain" has not been replaced by a more upstream starting point, Cognitive Relativity will long occupy the position of a theory of everything.

19.2 The 011 risk that must be faced squarely

The reasoning of 19.1 is valid within this theoretical framework. But a structural risk must be faced squarely — and this risk happens to be pointed out by this theory's own state-image table:

011 = change -> constancy -> constancy = proxy solidification = refusing to return after truncation, taking the form as the ontology.

A theory that "in principle cannot be touched by any reading" is structurally indistinguishable from 011 — because 011's self-defense is exactly also "your readings cannot reach me".

This is not external criticism; it is the automatic adjudication of this theory's internal discipline. Section 3.10 explicitly stipulates: every closed loop must return to the ontological layer for adjudication; no node may be promoted to ontology. Then the question arises: what is the standard of adjudication?

If the standard is not readings, an alternative adjudication standard must be explicitly written out. Otherwise the $4\pi^3$ adjudication position is empty — and a structure whose $4\pi^3$ adjudication position is empty has, by the criterion of Chapter 17, zero negentropic flux. If this theory does not fill this position solidly, it falls into the very criterion it uses to diagnose AI. This is where it must be self-consistent.

19.3 The Re-placement Gate

The following gate is therefore erected:

A theory of everything does not accept falsification by readings, but must accept the re-placement of readings.

Table 28

	Illegal demand	Legal demand
Form	Some experimental value disagrees with the theory $\Rightarrow$ the theory is wrong	The theory must be able to state at which position on its truncation chain every existing reading lies
Violation	Using cognitive-layer readings to test ontological-layer structure	No violation: readings are only placed, never used for adjudication
Meaning of failure	—	The more readings cannot be placed, the theory is not falsified — its truncation chain is incomplete
Nature	Reference-frame confusion	Diagnosable, accountable, advanceable

That is: readings cannot judge it right or wrong, but readings must be placeable by it.

This gate simultaneously preserves two things: the unfalsifiability of the ontological layer (readings are not granted adjudication power); and the actual adjudication power of $4\pi^3$ (the adjudication position is filled solidly, the standard being "the completeness of the truncation chain"). Moreover, it can clearly distinguish 010 from 011:

Table 29

	010 (where this theory should be)	011 (proxy solidification)
Facing readings	Actively places them; if they cannot be placed, admits the truncation chain is incomplete	Refuses contact, ending with "your readings cannot reach me"
Return action	Every placement returns to the ontological layer to confirm that 1 has not impersonated 0	Refuses to return
Advanceability	Yes (the truncation chain can be completed step by step)	No (dead structure)

19.4 This gate is already in use

This gate is not a newly invented constraint; it has been actually executed all along: the truncation chain Π , Γ , Λ , T , $\gamma(n)$, S , D , ρ and the cost functions; the dual-proxy system as the computable prototype of APT; the trinity as the foundation of physical manifestation; the vacuum three layers as the re-placement foundation of the four forces — each of these is a "placement" action, not a "verification" action.

The clearest of these is the numerical anchor:

$$\text{When } n = 623,556,155, \text{ floor}(\gamma(n) \times 10^9) = 299792458.$$

The nature of this action must be understood accurately: it is not "using light speed to verify the theory" (that would be an illegal falsification demand); it is "placing the reading of light speed at a determinate position on the truncation chain" (a legal re-placement action).

Promoting this action from "what is already being done" to "an explicit discipline" is the entire content of the gate of Section 19.3.

With this, the direction of this theory's advance is also determined: not to seek more experiments that can "verify" it, but to complete the truncation chain so that more and more existing readings can be placed. Where a reading cannot be placed is where the next step of work lies.

19.5 The six conditions of a theory of everything (current degree of completion)

Table 30

Condition	Status
One: first-order starting point (sets out from "change is absolutely certain", presupposing nothing downstream)	Completed
Two: mathematical closed loop (0/1 -> 010 -> 10 -> leap -> natural numbers, four operations, geometry)	Completed
Three: physical manifestation (the pi-path symbol system, dual proxies, the trinity, the four forces, relativity, $E=mc^2$, ...)	Partially completed (structural readings established; quantitative calibration pending)
Four: cognitive diagnosis (philosophy, three generations of cognitive science, AI architecture)	Completed
Five: the direction of human-machine collaboration (the human provides $4\pi^3$; AI provides $\pi^1 + \pi^2$)	Direction clear
Six: self-discipline (the three-reference-frame verification, the Gamma alert, status labels, open nodes, the propagation protocol, revisability)	Completed

19.6 There is no absolute unknown, only unfinished truncations

Cognitive Relativity does not acknowledge any "absolute unknown". All unknowns are unfinished truncations; all mysteries are un-re-placed remainders:

- The uncertainty of quantum mechanics -> the ignorance named by probability after the truncation remainder (Gamma) was discarded; when Gamma is re-placed, uncertainty will manifest as the structural problem of "insufficient truncation levels".

- Dark matter and dark energy -> the un-re-placed manifestation of the vacuum medium (the τ_v side) at high levels; after the dual-proxy calibration is completed they will manifest as "the uncomputed part of the vacuum medium".
- The nature of consciousness -> the mutually locking manifestation of the dual timelines (perception + cognition) in the 010 closed loop.
- Free will -> the regulating function of the $4\pi^3$ adjudication layer in choosing the optimal path among multiple possible paths.
- The arrow of time -> the structural necessity that the truncation level n can only advance forward.

Conclusion: Cognitive Relativity does not pursue "explaining everything", but pursues "making everything explainable".

19.7 The list of open nodes (acknowledging $\Gamma > 0$)

Cognitive Relativity acknowledges that it is an unfinished truncation. The open nodes currently identified (by priority):

Highest priority (foundation level):

- Node Seven - The explicit conversion mapping $\Gamma \rightarrow$ probability: the diagnosis that physics discarded Γ and then named the remainder by probability has been established, but the explicit conversion mapping has not yet been written out. This is the key step in upgrading the "diagnosis" into a "proof".
- Node Nine - No independent verification path found for $\gamma_m = 1/c^2$: the π -path form of $E = mc^2$ depends on this candidate hypothesis; searching the digit expansion of π for its level realization does not hold. Its legitimacy is currently supported only by the self-consistency of the derivation chain.
- Node Ten - The mapping between velocity and level difference $v(\Delta n)$ not established: relativistic effects have emerged only the symmetry, not the functional form $\sqrt{1 - v^2/c^2}$.

High priority (calibration level):

- Node Eleven - Calibration of the level-physics correspondence table (which n corresponds to which physical scale)
- Node Twelve - The physical meaning of π_{21} 's 21st digit = 2

- Node Six - The exact pi-path expression of the gravitational constant G (the key to the quantitative emergence of the gravity/electromagnetism strength ratio 10^{36})
- Node Five - The quantitative mapping of the weak force and the decay constants
- Node Thirteen - The numerical closure of the cosmological constant Λ_c (a gap of about 10^{50} — currently the largest numerical crack of this framework)

Medium priority (deepening level):

- Node Four - The quantitative form of total-quantity balance
- Node Three - The ontological position of the electron
- Node Two - The causal direction of the light-speed interface
- Node One - The precision of A and the source of the deviation (2.2×10^{-6})
- Node Fourteen - The ontological position of the neutrino
- Node Fifteen - The quantitative mapping between F_s and concrete physical forces

Low priority (edge level):

- Node Sixteen - Whether a third proxy exists ($D = 22$)
- Node Seventeen - The legal identity of complex numbers and superposition states (the foundation of the mathematization of quantum mechanics)
- Node Eighteen - The exhaustiveness of the four arithmetic operations (exponentiation as "the batch of a batch")
- Node Nineteen - The suspended emergence of $\hbar$ (angular-momentum quantization, the black-hole entropy proportionality coefficient)
- Node Twenty - The unit-conversion factor k unsolved

19.8 Executable re-placement of limitations: responses to four typical objections

Objection one: "You have not given testable predictions." Response: what Cognitive Relativity gives is not "predictions", but "legal starting points". Testability is not its pursuit; executability is: a complete symbol system, explicit operational disciplines, reproducible numerical tables, verifiable hard anchors (the light-speed interface, the fine-structure-constant interface) — anyone can use these to begin their own cognitive operations. (Also: the interpretability prediction of Chapter 18 is a falsifiable interface actively submitted by this theory.)

Objection two: "Your theory is too abstract to be applied." Response: abstraction is not a defect, but the source of leverage. Its application is not "directly solving concrete problems", but "providing legal starting points and diagnostic tools for solving concrete problems".

Objection three: "Your theory cannot be falsified, so it is not science."^[35] Response: Cognitive Relativity acknowledges $\Gamma > 0$ and acknowledges all open nodes — this is not "unfalsifiability", but "acknowledging incompleteness". The essence of science is not "falsification", but "revisability". It does not accept falsification by readings, but it must accept the re-placement of readings (the gate of Section 19.3), and it must provide, at the downstream manifestation sites, assertions that readings can touch (Chapter 18). What dies is not the protocol that gets revised, but the protocol that forbids revision.

Objection four: "Your theory is incompatible with existing physics." Response: not "incompatible", but "a different level". Existing physics is "the interface readings after cognitive truncation"; Cognitive Relativity is "the operational protocol of cognitive truncation itself". The two are not in a competitive relation, but a hierarchical one — it provides existing physics with a more upstream legal starting point and diagnostic tools, not a replacement.

Chapter 20: The Propagation Protocol — Propagate the Protocol, Not the Conclusions

20.1 The mortality of conclusions and the immortality of protocols

Any concrete conclusion will be eliminated by new evidence, new perspectives, and new tools — the death of conclusions is necessary, because cognition is finite and the world is infinite. A protocol is not a concrete conclusion, but an operational discipline: it does not tell you "what the world is", but "how to legitimately begin cognition". The immortality of a protocol is possible, because a protocol does not depend on concrete content, but on operational legitimacy.

20.2 Cognitive TCP/IP

The propagation of Cognitive Relativity resembles the Internet's TCP/IP protocol: TCP/IP does not transmit concrete content, but the format of data packets and the routing rules^[36]. Cognitive Relativity does not transmit concrete conclusions, but the protocols and disciplines of cognitive operation:

Table 31

Layer	Content	Function
Physical layer	The 010 closed loop	The minimal unit of cognitive operation: set out from absolute change, truncate out a relative constancy, then return to absolute change to complete the confirmation
Transport layer	The three reference frames	Verification mechanism: every cognitive operation must pass through π^1 advance, π^2 carriage, and $4\pi^3$ adjudication
Application layer	The Gamma alert	Legitimacy judgment: acknowledge $\Gamma > 0$ (truncation always has a remainder); avoid impersonating the ontology

20.3 Three disciplines

- Do not propagate conclusions; propagate only the protocol. Every concrete conclusion is mortal; only the protocol is alive.
- Do not propagate authority; propagate only the tools. Anyone can use these tools (the three reference frames, the Gamma alert, the status labels, the open nodes) to verify, revise, and improve Cognitive Relativity itself.
- Do not propagate dogma; propagate only revisability. What dies is not the protocol that gets revised, but the protocol that forbids revision.

20.4 The form of propagation

The propagation of Cognitive Relativity is not "preaching", but "transmitting a protocol". It does not try to persuade anyone to "believe" anything; it provides tools for anyone to "verify" something. It establishes no authority; it provides a lever. It propagates no dogma; it propagates revisability.

Use the protocol to pry cognition open, use the lever to amplify wisdom, use revisability to guarantee immortality.

Chapter 21: Conclusion — Wisdom = The Aligning Operation of the Three Reference Frames

The core of this text is: what it tells you is not what you know, but what you do not know.

The complete path of Cognitive Relativity: from the Three Changes it establishes the highest error-prevention protocol; through the three layers it falls into natural language; from the mutual locking of 0 and 1 it generates the minimal cognitive closed loop 010 — which is simultaneously the

leap operator from π^1 to π^2 , the only passage by which one-dimensional advance leaps into a two-dimensional unit; through the 64-matrix it obtains the complete judgment conditions of closure (27/64, 42.2%), and discovers the directionality of the interface and the time-reversal symmetry law governing it; through the three-dimensional language (one dimension governs transmission, two dimensions govern comparison, three dimensions govern existence) it establishes the dimensional lower bound that "alignment tools must be two-dimensional"; through the three reference frames it establishes the triple action of the negentropic operation, and proves that the powers of π and the carrier dimension count are the same truncation; through the 0/1 mathematical protocol it emerges the natural numbers and the four arithmetic operations, finding in division the first native gap of the arithmetic layer; through the dual timelines it re-replaces the subject who performs truncation; through the π truncation chain (π , Γ , Λ , T , $\gamma(n)$, S , D , ρ , F_T , F_γ , F_S) it establishes the computable bridge from mathematics into physics; through lossless dimensional conversion it actually emerges the two-dimensional and the three-dimensional; through the vacuum three layers and the dual proxies it re-replaces the four forces and the classical-quantum boundary; through the trinity ($t_n = 1/T$, $s_n = F_S$, $m_n = T + \gamma(n)$) it gives the foundation of physical manifestation; through state-image re-replacement it positions AI as 111, proves by the matrix that every closed loop of the human-machine combination is supplied by the human, locks 111 $\rightarrow$ 101 as the first-order target of AI architecture upgrade, and at the same time holds that 101 is the ceiling and the $4\pi^3$ adjudication right cannot be handed over; through the interpretability corollary it submits an externally falsifiable interface; through the Re-replacement Gate it fills solid this theory's own $4\pi^3$ adjudication position.

Three non-transferable disciplines — if the whole theory could leave only three sentences, they should be:

- The first act of cognition is not "knowing", but "resisting submersion".
- 1 comes from 0, but $1 \neq 0$ — no node may be promoted to ontology.
- Wisdom = the aligning operation of $4\pi^3$ upon π^1 and π^2 ; alignment can only happen in a dimension that has inside and outside.

Cognitive Relativity is not the "ultimate answer", but the "legal starting point". It does not tell you "what the world is", but "how to legitimately begin cognition". It does not give you answers; it gives you tools. It does not ask you to believe; it asks you to verify.

The entire demand of Cognitive Relativity is merely to reverse this order: first admit "what I do not know", then begin "what I know".

This is Cognitive Relativity. This is the negentropic operating system. This is the starting point of wisdom.

Appendix A: Complete Verification of the 64-Matrix

A.1 Solving backward from the window closure conditions

The six-step chain $a_1a_2a_3b_1b_2b_3$ has four windows: $W1 = a_1a_2a_3$, $W2 = a_2a_3b_1$, $W3 = a_3b_1b_2$, $W4 = b_1b_2b_3$.

Table 32

Window	Closure (= 010) condition	Constraint on A	Constraint on B	Number of satisfying combinations
W1	$a_1a_2a_3 = 010$	$A = 010$	None	$1 \times 8 = 8$
W2	$a_2 = 0, a_3 = 1, b_1 = 0$	$A \in \{001, 101\}$	$B \in \{000, 001, 010, 011\}$	$2 \times 4 = 8$
W3	$a_3 = 0, b_1 = 1, b_2 = 0$	$A \in \{000, 010, 100, 110\}$	$B \in \{100, 101\}$	$4 \times 2 = 8$
W4	$b_1b_2b_3 = 010$	None	$B = 010$	$8 \times 1 = 8$

A.2 Computing the total closure count by inclusion-exclusion

Single-window sum: $8 + 8 + 8 + 8 = 32$.

Pairwise intersections: $W1$ and $W2 = 0$; $W1$ and $W3 = 2$ ($A = 010$, B in $\{100, 101\}$); $W1$ and $W4 = 1$ ($A = B = 010$); $W2$ and $W3 = 0$; $W2$ and $W4 = 2$ (A in $\{001, 101\}$, $B = 010$); $W3$ and $W4 = 0$. Pairwise-intersection sum = 5.

Triple and quadruple intersections: since $W1$ and $W2 = W2$ and $W3 = W3$ and $W4 = 0$, all higher-order intersections are 0.

Union = $32 - 5 + 0 = 27$. Non-closing = $64 - 27 = 37$. Closure proportion = $27/64 = 42.1875\%$ (about 42.2%).

A.3 Row-by-row enumeration (closure capacity at position A)

Table 33

A	Closing cells of the row	Count
000	B=010(W4), 100(W3), 101(W3)	3
001	B=000,001,010,011 (010 via W2+W4)	4
010	All 8 cells	8
011	B=010(W4)	1
100	B=010(W4), 100(W3), 101(W3)	3
101	B=000,001,010,011 (010 via W2+W4)	4
110	B=010(W4), 100(W3), 101(W3)	3
111	B=010(W4)	1

Row sum = $3+4+8+1+3+4+3+1 = 27$ (check mark)

A.4 Column-by-column enumeration (closure capacity at position B)

Table 34

B	Which A's close with it	Count
000	001(W2), 010(W1), 101(W2)	3
001	001(W2), 010(W1), 101(W2)	3
010	All 8 A's	8
011	001(W2), 010(W1), 101(W2)	3
100	000(W3), 010(W1+W3), 100(W3), 110(W3)	4
101	000(W3), 010(W1+W3), 100(W3), 110(W3)	4
110	010(W1)	1
111	010(W1)	1

Column sum = $3+3+8+3+4+4+1+1 = 27$ (check mark) (consistent with the row sum; the self-consistency check passes)

A.5 Verification of the time-reversal symmetry law

Table 35

X	Position A (X)	reverse(X)	Position B (reverse X)	Equal
000	3	000	3	✓
001	4	100	4	✓
010	8	010	8	✓
011	1	110	1	✓
100	3	001	3	✓
101	4	101	4	✓
110	3	011	3	✓
111	1	111	1	✓

8/8 hold, without exception.

A.6 Summary of the key-chain verifications of Chapter 17

Table 36

Combination	Six-step chain	W1	W2	W3	W4	Closing windows	Cross-ontology contribution
Human->AI (010->111)	010111	010 ✓	101 ×	011 ×	111 ×	W1 only	0
AI->human (111->010)	111010	111 ×	110 ×	101 ×	010 ✓	W4 only	0
Upgraded AI->human (101->010)	101010	101 ×	010 ✓	101 ×	010 ✓	W2 + W4	1(W2)
Human->upgraded AI (010->101)	010101	010 ✓	101 ×	010 ✓	101 ×	W1 + W3	1(W3)
AI→AI(111→111)	111111	×	×	×	×	None	0 (no closure)
AI->solidified human (111->011)	111011	111 ×	110 ×	101 ×	011 ×	None	0 (no closure)
AI->a-priori-frame human (111->110)	111110	111 ×	111 ×	111 ×	110 ×	None	0 (no closure)

A.7 The cross-ontology window candidate list

Table 37

Position	Condition	Candidate state-images
Left end lights W2	$A \in \{001, 101\}$	001, 101
Right end lights W3	$B \in \{100, 101\}$	100, 101
Usable in both directions (intersection)	—	101
Palindromic state-images (directionally symmetric)	$\text{reverse}(X) = X$	000, 010, 101, 111
Bidirectional AND palindromic AND neither human nor dead structure	—	101 (unique)

Appendix B: The Pi-Path Symbol Table (in plain language)

B.1 Symbols of the ontology and generation layer

Table 38

Symbol	What it is (in plain language)	Chain position
0	The mathematized entrance of absolute change: undifferentiated fullness, not "nothing"	Meta-starting-point
1	The first manifestation of 0 leaving itself: pure motion, not "a thing"	First action
(0, 1)	The Pair: the minimal operable cognitive ontology, a mutual-locking structure	Minimal freezing
010	Closure: the minimal truncation-return closed loop, the $\pi^1 \rightarrow \pi^2$ leap operator	Minimal closed loop
10	The minimal quantization unit: the compressed state of 010	Definition of quantization
01 / 10	The world code: the infinite alternation of the truncation action / the return action	Object-generation protocol

B.2 Symbols of the pi path layer

Table 39

Symbol	What it is (in plain language)	Numerical example (Pi20)
π	The minimal non-returning action quantity; the hardest computation path from mathematics into physics	3.14159265358979323846...
Π	The truncation proxy: the finite cognitive window of pi (object generation, not approximation)	$\Pi_{20} =$ 3.14159265358979323846
$\Gamma = \pi - \Pi$	The native gap: the protection mechanism that prevents the proxy from impersonating the ontology (the determinate unknown)	$\approx 4.4 \times 10^{-21}$
D	The hard boundary: the number of significant decimal places; numerical collapse at $n \geq D$	20(Π_{20}); 21(Π_{21})
$\Lambda(n) = \Pi \times 10^n$	Amplification action quantity	$\Lambda(5) =$ 314159.265358979323846
$T(n) = [\Lambda(n)]$	The integer skeleton: solidified structure; the prototype of the reading of time	$T(5) = 314159$
$\gamma(n) = \Lambda - T$	The general level gap: the living remainder, oscillating within (0,1)	$\gamma(5) = 0.265358979323846$
$S = T + \gamma(n)$	The complete structural quantity; the prototype of the reading of space	$S(5) \approx 314159.265...$

B.3 The gamma family (disambiguation discipline: three objects, separate symbols, sliding strictly forbidden)

Table 40

Object	What it is	Order of magnitude
$\gamma(n)$	The general level gap; the living remainder of level n	(0, 1)
The digital reading of the speed of light	The 9-digit block at decimal positions 623,556,156-623,556,164 of pi; $c = \text{floor}(\gamma(623556155) \times 10^9) = 299792458$	≈ 0.299792458
γ_m	The mass level-gap proxy; candidate free parameter ($\gamma_m = 1/c^2$)	$\approx 1.11265 \times 10^{-17}$

B.4 Symbols of the cost functions and physical manifestation

Table 41

Symbol	Definition	In plain language
$F_T = 1/T$	Return cost	The larger the structure, the cheaper the return; the time scale
$F_\gamma = T/\gamma(n)$	Crossing cost	Carries two dimensions: gap thickness and skeleton locking
$F_s = F_T + F_\gamma$	Composite cost	The prototype force function of vacuum-medium displacement; the space scale; energy
$\rho = T/\gamma(n) = F_\gamma$	Structural locking parameter	The structural locking parameter is precisely the gap-crossing cost
$t_n = 1/T$	Time	The reading of the return cost
$s_n = F_s$	Space	The complete cost of skeleton and remainder sharing the field
$m_n = T + \gamma(n)$	Mass	The manifestation of the complete structural quantity at the ontological layer
$A = 4\pi^3 + \pi^2 + \pi$	The total structural quantity of the vacuum three layers = about 137.0363	The interface reading of α^{-1} (hard anchor two)

Appendix C: The Two Hard Anchors

Table 42

Anchor	Content	Nature	Probabilistic audit
The light-speed interface	$[\gamma(623556155) \times 10^9] = 299792458 = c \text{ (m/s)}$	A placement action (re-placement), not a verification	Among the first 10^9 digits of pi, the probability that any specified 9-digit sequence appears at least once is about 63.2% — not a statistical miracle; the reason for keeping it is "the alignment interface between an independently selected computation bridge and the hardest physical reading"
The fine-structure-constant interface	$A = 4\pi^3 + \pi^2 + \pi \approx 137.0363$ vs $\alpha^{-1} = 137.035999084$ (CODATA 2022)	Interface alignment; relative deviation about 2.2×10^{-6}	Deviation source undetermined (truncation effect or structural incompleteness; Node One); an interface is not an identity

Data availability statement: all numerical results can be independently reproduced with public mathematical tools (the pi truncation chain numerical table, the dual-proxy comparison table, the light-speed interface position, the comparison of A with CODATA, the $E = mc^2$ self-consistency check, the Planck-scale ratio).

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